

Attention-aware task scheduling under partial observability of human cognitive state: a memory-augmented deep reinforcement learning approach

P. N. Bhargav Teja^a, Lanka Sree Chathurya^a, Rajay Vedaraj I.S.^{a,*}

^aVellore Institute of Technology, Vellore, 632014, Tamil Nadu, India

Abstract

Dispatching rules such as earliest-deadline-first (EDF) and shortest-processing-time (SPT) assume constant processing capacity, which fails when the resource is a person whose attention depletes, whose fatigue accumulates and whose alertness follows a circadian rhythm. We formulate human-centric task scheduling as a partially observable Markov decision process in which rest is an explicit action, attention and fatigue are latent states observed through noisy estimates, and a multi-objective reward balances deadline-weighted task value against cognitive load. We propose an attention-aware deep Q-network (AA-DQN) that combines an observation history, as an approximation of the belief state, with Double Q-learning and a dueling architecture. In a cognitively grounded simulator, AA-DQN was benchmarked against nine rule-based schedulers, including grid-tuned EDF and SPT rules with periodic or state-triggered rest, using ten training seeds, 500 held-out workdays with common random numbers and multiplicity-corrected tests. Rules that never rest performed poorly (EDF: 46.8% of tasks on time). The pre-specified AA-DQN matched the return of the best tuned rule and completed more tasks on time (72.8% versus 71.4%). A compact variant with intensity-only

*Corresponding author.

Email address: rajay@vit.ac.in (Rajay Vedaraj I.S.)

actions, evaluated as a secondary analysis, exceeded the best rule in return (+0.31, $p < 0.001$) and on-time completion (74.0%). The learned policies alone combined high throughput with protection of demanding tasks: they front-load hard work while attention is high and rest around the post-lunch dip. Their advantage grew markedly with accurate cognitive-state estimates, and they were robust to sensor noise, whereas generalisation to unseen workloads remained limited.

7 *Keywords:* Human-centric scheduling, Deep reinforcement learning, Partially
8 observable Markov decision process, Cognitive fatigue, Attention, Rest-break
9 scheduling, Industry 5.0

10 **1. Introduction**

11 Scheduling decides which piece of work is executed when, and it is one of the
12 oldest problems in operations research and computer engineering [1–4]. Classical
13 dispatching rules such as First-In-First-Out (FIFO), Shortest Processing Time
14 (SPT) and Earliest Deadline First (EDF) are attractive because they are transparent,
15 cheap to compute and, under the right assumptions, provably optimal: EDF
16 minimises maximum lateness on a single resource with constant capacity [2, 5],
17 and SPT minimises mean flow time [1]. The common assumption underlying these
18 guarantees is that the resource executing the work has a *fixed, state-independent*
19 *processing capacity*. This assumption is reasonable for processors and machines
20 but is violated whenever the resource is a person.

21 Human work capacity is dynamic. Sustained effort depletes attentional re-
22 sources and produces the vigilance decrement first documented by Mackworth [6]
23 and since replicated across many tasks [7–9]; mental fatigue accumulates with
24 time-on-task and degrades both speed and accuracy [10–13]; and alertness follows

25 a circadian rhythm with a pronounced early-afternoon dip [14–17]. Rest breaks
26 partially reverse these effects, and field studies consistently report that well-timed
27 breaks preserve or even increase productivity despite the time they consume [18–
28 22]. A schedule that is optimal for a constant-capacity machine can therefore be
29 poor for a human: it front-loads demanding work regardless of cognitive state,
30 never allocates recovery, and drives the worker into a low-capacity regime in
31 which deadlines are missed precisely because capacity was ignored. The concern
32 is not only productivity. Chronic fatigue is associated with errors and accidents
33 [23, 24], and the move from Industry 4.0 towards a human-centric Industry 5.0
34 explicitly calls for production and service systems that adapt to the well-being of
35 their workers rather than the reverse [25–29].

36 Operations-management research has long recognised this gap. Models of
37 worker fatigue and recovery have been integrated into dual-resource-constrained
38 systems [30–32], rest-allowance and break-scheduling problems have been formu-
39 lated analytically [33–35], and taxonomies for merging scheduling theory with
40 human factors have been proposed [36–39]. These approaches, however, usually
41 assume that the worker’s state is known or deterministic and optimise an offline
42 plan; they do not provide an adaptive *policy* that reacts to the evolving, uncertain
43 state of a specific person during the day. Deep reinforcement learning (DRL),
44 in turn, has produced effective adaptive schedulers for job shops, clusters and
45 production lines [40–46], and recent surveys document rapid growth of the field
46 [47–49]. Yet the resources in these studies are machines, and the few recent studies
47 that include worker fatigue embed it in large shop-floor models where fatigue is a
48 deterministic, fully observed variable [50, 51]. Two features of the human problem
49 are thus rarely addressed together: *the decision of when to rest is itself a scheduling*

50 *action, and the cognitive state that should drive that decision cannot be measured*
51 *directly*. In practice, attention and fatigue are inferred from noisy proxies such
52 as physiological signals, eye tracking, performance probes or self-report [52–56],
53 which makes the scheduling problem partially observable.

54 This paper studies attention-aware task scheduling for a single knowledge or
55 production worker as a partially observable Markov decision process (POMDP)
56 [57–59] and solves it with a memory-augmented deep Q-network. We pay par-
57 ticular attention to the methodological issues that make DRL comparisons hard
58 to trust [60, 61]: we compare against strong, *tuned*, cognition-aware heuristics
59 rather than only against FIFO and EDF; we use disjoint seed sets for training,
60 tuning and testing; we evaluate on 500 held-out workdays with common random
61 numbers; we report ten independent training seeds with confidence intervals and
62 multiplicity-corrected paired tests; and we separate effects of the reward function
63 from effects of the dynamics through ablation.

64 The contributions are as follows.

- 65 (i) **A cognitively grounded scheduling simulator.** We formulate a workday-
66 scale scheduling environment in which a latent homeostatic attention re-
67 source, an exponentially accumulating and recovering fatigue state [30],
68 and a circadian process [14, 15] jointly determine the worker’s instanta-
69 neous work rate. Tasks arrive stochastically with heterogeneous difficulty,
70 work content and deadlines. Rest is an explicit action. The simulator, its
71 parameters and their behavioural rationale are fully specified and released.
- 72 (ii) **A POMDP formulation with a multi-objective reward.** The scheduler ob-
73 serves only noisy estimates of attention and fatigue together with a compact,
74 size-invariant summary of the task queue. The reward combines deadline-

75 weighted task value with explicit attention and fatigue terms, and we analyse
76 the resulting productivity–fatigue trade-off as a Pareto front [62, 63].

77 (iii) **An attention-aware DQN (AA-DQN).** We combine a finite observation
78 history, which acts as an approximate sufficient statistic for the belief state
79 [64–66], with Double Q-learning [67] and a dueling architecture [68]. A
80 component ablation quantifies the contribution of each element.

81 (iv) **A rigorous empirical study.** Against nine rule-based baselines, including
82 grid-tuned heuristics that combine EDF or SPT sequencing with periodic
83 or state-triggered rest and observe the same noisy cognitive signals, we test
84 (a) whether learning helps, (b) *why* it helps, through environment ablations
85 that remove observation noise, the circadian process, the cognitive dynamics
86 or the cognitive reward terms, (c) what behaviour is learned, and (d) how
87 the learned policy generalises to unseen worker profiles, workloads and
88 sensor-noise levels.

89 The remainder of the paper is organised as follows. Section 2 reviews related
90 work and positions the study. Section 3 formulates the problem. Section 4 presents
91 AA-DQN. Section 5 details the experimental protocol. Section 6 reports results,
92 Section 7 discusses implications, ethics and limitations, and Section 8 concludes.

93 **2. Related work**

94 *2.1. Classical and dynamic scheduling*

95 Single-resource sequencing theory provides the canonical dispatching rules
96 and their optimality conditions: SPT for mean flow time [1], EDF for maximum
97 lateness [2], and the Moore–Hodgson algorithm for the number of late jobs [69].

98 The three-field classification of Graham et al. [3] and the textbook treatment of
99 Pinedo [4] organise the deterministic theory. In real-time computing, EDF is
100 optimal for preemptive scheduling of independent tasks whenever the processor
101 is not overloaded [5], but its performance can collapse under overload, when
102 a domino effect causes many deadlines to be missed [70]. This failure mode is
103 directly relevant here, because a fatigued worker is in effect an overloaded resource.
104 Surveys of priority rules [71, 72] and of dynamic and reactive scheduling [73, 74]
105 show that practice relies heavily on such rules when information arrives online. In
106 all of these models, however, processing times are exogenous: they do not depend
107 on the history of the schedule.

108 2.2. *Human factors in scheduling and workforce models*

109 Personnel scheduling and rostering [75–77] assign workers to shifts but typ-
110 ically treat performance within a shift as constant. Research at the interface of
111 operations and human resources argues that this abstraction misses first-order
112 effects [37, 38], and Lodree et al. [36] proposed a taxonomy for integrating hu-
113 man factors into scheduling theory. Quantitative models of learning, forgetting,
114 fatigue and recovery have been developed for dual-resource-constrained systems
115 [30–32, 78]. They use exponential fatigue accumulation during work and expo-
116 nential recovery during rest, which is the functional form we adopt. Rest-break
117 scheduling has been formulated as an optimisation problem since Bechtold et
118 al. [33], with ergonomic guidance on work–rest ratios summarised by Konz [34]
119 and rest-allowance models for task assignment developed by Calzavara et al. [35].
120 Fatigue has also been incorporated into flexible job-shop formulations solved by
121 metaheuristics [50] and, very recently, by graph-based DRL [51]. Reviews of hu-
122 man factors in Industry 4.0 and in order picking [39, 79–81] conclude that worker

123 heterogeneity and well-being are still under-represented in operational models. A
124 common feature of this literature is that the worker state is assumed to be known,
125 either deterministic or computed from the schedule, so the decision problem is
126 fully observable.

127 *2.3. Deep reinforcement learning for scheduling*

128 Reinforcement learning has been applied to scheduling since the mid-1990s
129 [82, 83], and the combination of Q-learning [84] with deep function approximation
130 [85] enabled scheduling policies that operate on rich state representations. Notable
131 examples include cluster resource management [40], data-processing job schedul-
132 ing with graph neural networks [41], semiconductor production scheduling [42],
133 dynamic flexible job shops with job insertions [43, 46], and size-agnostic job-shop
134 dispatching [44, 45, 86]. Parameter-adaptive RL for dynamic job shops [87] and
135 RL for combinatorial optimisation more broadly [88, 89] complete the picture.
136 Systematic reviews [47–49] note three recurring weaknesses: comparisons against
137 untuned or weak baselines, limited statistical analysis, and little attention to human
138 resources. Our study addresses all three.

139 *2.4. Cognitive state, its measurement, and adaptive systems*

140 Resource theories of attention [90, 91] and cognitive load theory [92] model
141 mental capacity as a limited resource. Vigilance research shows that this resource
142 is depleted by sustained effort [6, 7] and partially restored by brief breaks [93].
143 Accounts of fatigue as an effort-regulation or opportunity-cost signal [12, 94]
144 and evidence linking fatigue to disengagement [95] motivate treating fatigue as a
145 state variable that shapes future capacity. Biomathematical fatigue models, from
146 the two-process model of sleep regulation [96–98] to the three-process model of

147 alertness [14] and applied models such as SAFTE [99], combine homeostatic and
148 circadian components. We borrow this decomposition at the time scale of a single
149 workday. Recovery research in organisational psychology confirms the value of
150 micro-breaks [22, 100–102]. On the measurement side, workload and fatigue can
151 be estimated from EEG, cardiac, ocular and behavioural signals [52–54, 103, 104],
152 from vigilance probes such as the PVT [55, 105], or from questionnaires such as
153 NASA-TLX [56, 106]. All of these estimates are noisy. Adaptive automation
154 and physiological computing use such estimates to adjust system behaviour [107–
155 112], and interruption-management systems decide when to notify a user based
156 on inferred cognitive state [113–116]. Reinforcement learning has been used
157 to personalise just-in-time health interventions [117–120], and in human–robot
158 collaboration to adapt task allocation to human capability and fatigue [121–125].
159 These works show that cognitive-state estimates can drive decisions, but they rarely
160 address the sequential scheduling of a task queue under deadlines.

161 2.5. *Partial observability and multiple objectives in RL*

162 When the state is not directly observed, the optimal policy is a function of the
163 belief state [57–59]. Memoryless policies can be arbitrarily poor [64]. Practical
164 deep RL approximates the belief either with recurrent networks [65, 126, 127]
165 or with finite observation histories, as in the frame stacking of Mnih et al. [85].
166 Ni et al. [66] show that well-tuned memory-based model-free agents are strong
167 POMDP baselines. Stabilisation techniques for DQN, namely experience replay
168 [128], target networks [85], Double Q-learning [67], dueling heads [68] and their
169 combinations [129], are well established, as is the theory of TD learning with func-
170 tion approximation [130]. Scheduling for humans is inherently multi-objective:
171 productivity, deadlines and well-being conflict. Multi-objective RL provides the

172 vocabulary of scalarisation and Pareto fronts [62, 63, 131]. Reward shaping must
173 be used with care, because additional reward terms can change the optimal policy
174 [132]. Finally, the deployment of RL in settings that involve people raises safety
175 and reliability concerns [133–135], which we address in Section 7.

176 2.6. *Research gap*

177 Table 1 positions this work against representative streams. To our knowledge,
178 no prior study simultaneously (i) models dynamic attention, fatigue and circadian
179 variation as endogenous state, (ii) treats rest as a scheduling action, (iii) assumes
180 that the cognitive state is only partially observable, (iv) learns an adaptive policy,
181 and (v) benchmarks that policy against tuned, cognition-aware heuristics with
182 multi-seed statistical testing. Our study fills this gap.

Table 1: Positioning of this study relative to representative research streams (✓: addressed; ◦: partially or in a simplified form; –: not addressed).

Stream (representative works)	Dynamic human state	Rest as action	Partial observ.	Adaptive policy	Deadlines/ queue	Tuned cog. baselines
Dispatching rules [1, 2, 5, 72]	–	–	–	–	✓	–
Fatigue/recovery models in DRC systems [30–32]	✓	◦	–	–	◦	–
Rest-break / rest-allowance optimisation [33–35]	◦	✓	–	–	–	–
DRL for machine scheduling [40, 42–44, 46]	–	–	–	✓	✓	–
Fatigue-aware shop schedul- ing [50, 51]	✓	–	–	◦	✓	–
Adaptive automation / inter- ruption mgmt. [107, 112– 114]	✓	◦	✓	◦	–	–
This study	✓	✓	✓	✓	✓	✓

3. Problem formulation

We consider a single worker who processes a stream of tasks during one work-day. A scheduler, which could be a digital assistant, a workflow-management system or a supervisory controller in a human–robot cell, decides at regular intervals what the worker should do next: start or continue an easy task, a hard task, or take a short break. The worker’s capacity depends on a latent cognitive state that evolves with the schedule. Table 2 summarises the notation and Table 3 lists the

190 parameter values.

Table 2: Notation.

Symbol	Meaning
$t \in \{0, \dots, T-1\}, \Delta$	decision slot index and slot length ($T = 48, \Delta = 10$ min)
h_t	clock time of slot t (h), $h_t = 9 + t\Delta/60$
a_i, d_i, w_i, D_i	arrival slot, difficulty, work content and deadline of task i
H_t	homeostatic attention resource (latent)
$c(h), A_t$	circadian modulation and effective attention, $A_t = \text{clip}(H_t + \gamma c(h_t))$
F_t	fatigue (latent)
ρ_t	work rate (work units completed per slot)
$\tilde{A}_t, \tilde{F}_t$	noisy observations of A_t, F_t
o_t, x_t	observation vector and history window of length k
u_t, b_t	indicators of working and resting in slot t
r_t	scalar reward; λ, μ fatigue and attention weights
γ_{RL}	RL discount factor (distinct from circadian amplitude γ)

191 3.1. Workload model

192 The day is divided into $T = 48$ slots of $\Delta = 10$ minutes (09:00–17:00). Each
 193 day contains $N = 12$ tasks. Six tasks are available at the start of the day and
 194 the remaining six arrive at slots drawn uniformly from $\{1, \dots, \lfloor 0.6T \rfloor\}$, which
 195 models interruptions and new requests. Task i has difficulty $d_i \sim \mathcal{U}[0.2, 1.0]$ and
 196 is labelled *hard* if $d_i \geq 0.6$ and *easy* otherwise. Its work content is $w_i = 1 + 2.5 d_i$
 197 work units, so a task requires between 1.5 and 3.5 slots (15–35 min) at full capacity.
 198 Its deadline is $D_i = \min\{T, a_i + \lceil s_i T \rceil\}$ with relative slack $s_i \sim \mathcal{U}[0.3, 1.0]$. With

199 these settings, the expected daily work content (≈ 30 units) is below the capacity
 200 of a fully rested worker (48 units) but above what a worker can deliver once
 201 attention and fatigue degrade. Scheduling decisions therefore matter: the day is
 202 *conditionally overloaded*.

203 3.2. Cognitive dynamics

204 *Effective attention.* Following the decomposition of alertness into homeostatic and
 205 circadian components [14, 96, 98], effective attention is

$$A_t = \text{clip}(H_t + \gamma c(h_t), 0, 1), \quad c(h) = 0.2 e^{-(h-10.5)^2/(2 \cdot 1.5^2)} - e^{-(h-14)^2/(2 \cdot 1.2^2)}, \quad (1)$$

206 where $c(h)$ encodes a late-morning peak and the post-lunch dip around 14:00
 207 [15–17] (Fig. 1a), and $\gamma = 0.15$ sets its amplitude.

208 *Homeostatic attention resource.* Working on a task of difficulty d depletes the
 209 attentional resource, and the depletion is accelerated by fatigue. This captures the
 210 vigilance decrement and resource-control accounts of sustained attention [6–8].
 211 Resting restores the resource exponentially towards its maximum [22, 93]:

$$H_{t+1} = \text{clip}\left(H_t - \alpha d (1 + \phi F_t) u_t + \beta (1 - H_t) b_t + \sigma_p \xi_t, 0, 1\right), \quad \xi_t \sim \mathcal{N}(0, 1), \quad (2)$$

212 where $u_t, b_t \in \{0, 1\}$ indicate work and rest ($u_t + b_t = 1$), $\alpha = 0.02$ is the depletion
 213 rate per unit difficulty, $\phi = 0.5$ the fatigue coupling, $\beta = 0.30$ the per-slot recovery
 214 fraction, and $\sigma_p = 0.02$ the process-noise scale that represents unmodelled day-
 215 to-day and moment-to-moment variability.

216 *Fatigue.* Fatigue accumulates exponentially towards saturation during work and
 217 decays exponentially during rest. This is the discrete-time analogue of the fatigue–

218 recovery model of Jaber and Neumann [30], also used in [31, 32]:

$$F_{t+1} = \text{clip}\left(F_t + \delta d (1 - F_t) u_t - \eta F_t b_t, 0, 1\right), \quad (3)$$

219 with accumulation rate $\delta = 0.05$ and recovery rate $\eta = 0.20$.

220 *Productivity.* While working on task i , the worker completes

$$\rho_t = \rho_0 A_t^{0.5+d_i} (1 - \kappa F_t) \quad (4)$$

221 work units in slot t , with $\rho_0 = 1$ and $\kappa = 0.5$. The exponent makes hard tasks more
 222 sensitive to attention than easy ones (Fig. S1 in the Supplementary Material), in
 223 line with resource theories in which demanding tasks draw more heavily on limited
 224 capacity [90, 91]. The fatigue factor reduces throughput multiplicatively [10, 12].
 225 A task is complete at the end of the first slot in which its cumulative work reaches
 226 w_i .

227 *Initial state and behavioural plausibility.* Each day starts with $H_0 \sim \mathcal{U}[0.8, 1.0]$
 228 and $F_0 \sim \mathcal{U}[0, 0.15]$. Parameters were chosen *a priori* to reproduce qualitative
 229 patterns reported in the time-on-task and fatigue literature rather than fitted to a
 230 particular dataset. Under continuous work without breaks, attention falls from
 231 ≈ 1.0 to ≈ 0.3 by the end of the day, with a visible early-afternoon trough, and
 232 fatigue rises to ≈ 0.85 . One 10-minute break recovers 30% of the attentional deficit
 233 and 20% of accumulated fatigue (Fig. 1b,c). Because no single parameterisation
 234 can represent all workers, Section 6.8 evaluates the policies zero-shot on resilient
 235 and fatigue-prone profiles in which all four rate parameters are scaled by $\pm 30\%$.

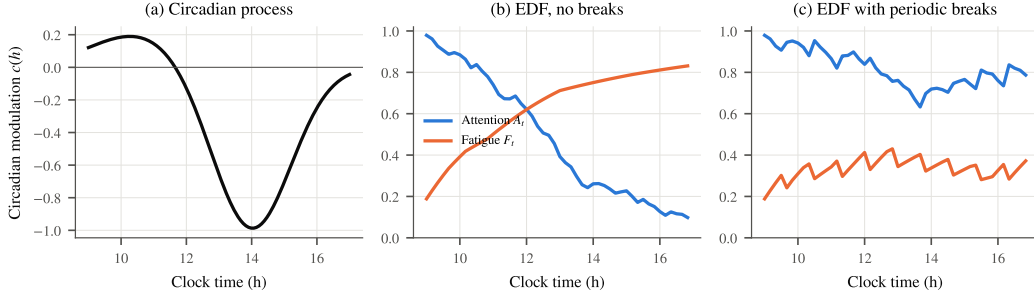


Figure 1: Simulator dynamics. (a) Circadian modulation $c(h)$ with a late-morning peak and post-lunch dip. (b) Effective attention and fatigue on one held-out test day under EDF without breaks. (c) The same day under EDF with a break after every four work slots (shaded bands mark breaks). Breaks keep attention high and fatigue low at the cost of working time.

3.3. Observation model and partial observability

The scheduler cannot observe H_t , A_t or F_t . It receives noisy estimates

$$\tilde{A}_t = \text{clip}(A_t + \sigma_o \epsilon_t^A, 0, 1), \quad \tilde{F}_t = \text{clip}(F_t + \sigma_o \epsilon_t^F, 0, 1), \quad \epsilon_t^A, \epsilon_t^F \sim \mathcal{N}(0, 1), \quad (5)$$

with $\sigma_o = 0.10$ in the nominal setting. These estimates stand in for wearable, ocular, behavioural or self-report estimators, whose errors are substantial in practice [52–54]. The full observation is the 13-dimensional vector

$$o_t = \left[\frac{t}{T}, \tilde{A}_t, \tilde{F}_t, \underbrace{\frac{n_t^e}{N}, \frac{W_t^e}{20}, \sigma_t^e, \frac{m_t^e}{N}, \frac{\omega_t^e}{4}}_{\text{easy class}}, \underbrace{\frac{n_t^h}{N}, \frac{W_t^h}{20}, \sigma_t^h, \frac{m_t^h}{N}, \frac{\omega_t^h}{4}}_{\text{hard class}} \right], \quad (6)$$

where, for each class, n_t is the number of available unfinished tasks, W_t their remaining work, $\sigma_t = \text{clip}((\min_i D_i - t)/T, -1, 1)$ the normalised slack of the most urgent task (1 if the class is empty), m_t the number of overdue tasks, and ω_t the remaining work of the shortest task (0 if the class is empty). Unlike a binary completion vector, this summary is invariant to the number of tasks and exposes the deadline information that urgency-aware scheduling requires. The worker’s

247 true state and the future arrivals are hidden, so the decision problem is a POMDP
 248 $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma_{\text{RL}} \rangle$ [59]. Here $\mathcal{S}$ contains the full task set and (H_t, F_t, t) , Ω
 249 is the observation space of Eq. (6), O is given by Eq. (5), and P by Eqs. (1)–(4)
 250 together with the arrival process.

251 3.4. Action space

252 The agent chooses among five macro-actions,

$$\mathcal{A} = \{\text{EASY}, \text{HARD}, \text{EDF}, \text{SPT}, \text{BREAK}\}. \quad (7)$$

253 EASY (HARD) dispatches the available easy (hard) task with the earliest deadline,
 254 falling back to the other class if the chosen class is empty. EDF and SPT dispatch,
 255 across both classes, the task with the earliest deadline or the least remaining work.
 256 BREAK allocates the slot to rest, and the worker also rests when no task is available.
 257 The first two actions let the agent regulate the cognitive *intensity* of work relative
 258 to the worker’s state. The two rule actions let it switch between urgency-driven and
 259 throughput-driven sequencing, in the spirit of DRL schedulers that select among
 260 dispatching rules [43, 48]. This hybrid design has an important methodological
 261 consequence: every rule-based baseline in Section 5.1 can be expressed as a
 262 policy over $\mathcal{A}$, so a learned policy is never handicapped relative to the baselines
 263 by its action space. Section 6.6 reports an agent restricted to the intensity actions
 264 $\{\text{EASY}, \text{HARD}, \text{BREAK}\}$ of the original formulation. The action space is independent
 265 of N , and tasks are preemptible at slot boundaries, which is common in knowledge
 266 work.

267 3.5. Reward

268 The per-slot reward combines task outcomes with cognitive terms:

$$r_t = \underbrace{\sum_{i \in C_t} (1 + d_i) [\mathbb{I}(t+1 \leq D_i) + 0.25 \mathbb{I}(t+1 > D_i)] - 0.5 |\mathcal{M}_t|}_{r_t^{\text{task}}} + \mu A_{t+1} - \lambda F_{t+1}, \quad (8)$$

269 where C_t is the set of tasks completed in slot t , $\mathcal{M}_t$ is the set of tasks whose deadline
 270 expires unfinished at the end of slot t , $\mu = 0.05$ and $\lambda = 0.10$. Harder tasks are
 271 worth more, late completion retains a quarter of the value, and each missed deadline
 272 costs 0.5. The attention and fatigue terms express the organisation’s preference for
 273 sustainable work. Because such terms can change the optimal policy [132], we (i)
 274 report reward-independent operational metrics (on-time rate, completion, fatigue
 275 exposure) alongside the return, (ii) train an agent with $\lambda = \mu = 0$ to show that
 276 the learned behaviour is driven by the dynamics rather than by the reward terms,
 277 and (iii) sweep λ to trace the productivity–fatigue Pareto front. The objective
 278 is to find a policy π that maps observation histories to actions and maximises
 279 $\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma_{\text{RL}}^t r_t \right]$.

Table 3: Nominal simulator parameters and their rationale.

Parameter	Value	Rationale / source
Slots per day T , slot length Δ	48, 10 min	8-h workday; decision granularity of micro-breaks [22]
Tasks per day N (initial / arriving)	12 (6 / 6)	conditionally overloaded day; stochastic arrivals
Difficulty d_i , hard threshold	$\mathcal{U}[0.2, 1]$, 0.6	range of the original formulation
Work content w_i	$1 + 2.5d_i$ units	15–35 min at full capacity
Relative slack s_i	$\mathcal{U}[0.3, 1]$	range of the original formulation
Attention depletion α , fatigue coupling ϕ	0.02, 0.5	vigilance decrement over hours of work [7, 8]
Attention recovery β	0.30 per slot	break benefits [22, 93]
Circadian amplitude γ	0.15	post-lunch dip [15, 16]
Fatigue accumulation δ , recovery η	0.05, 0.20	exponential fatigue–recovery [30, 31]
Fatigue throughput penalty κ	0.5	performance loss under fatigue [10, 136]
Process / observation noise σ_p, σ_o	0.02, 0.10	variability; sensor error [52, 54]
Reward weights μ, λ	0.05, 0.10	swept in Section 6.5

280 4. Attention-aware deep Q-network

281 Figure 2 gives an overview of the closed loop between the simulated workday
 282 and the learning agent.

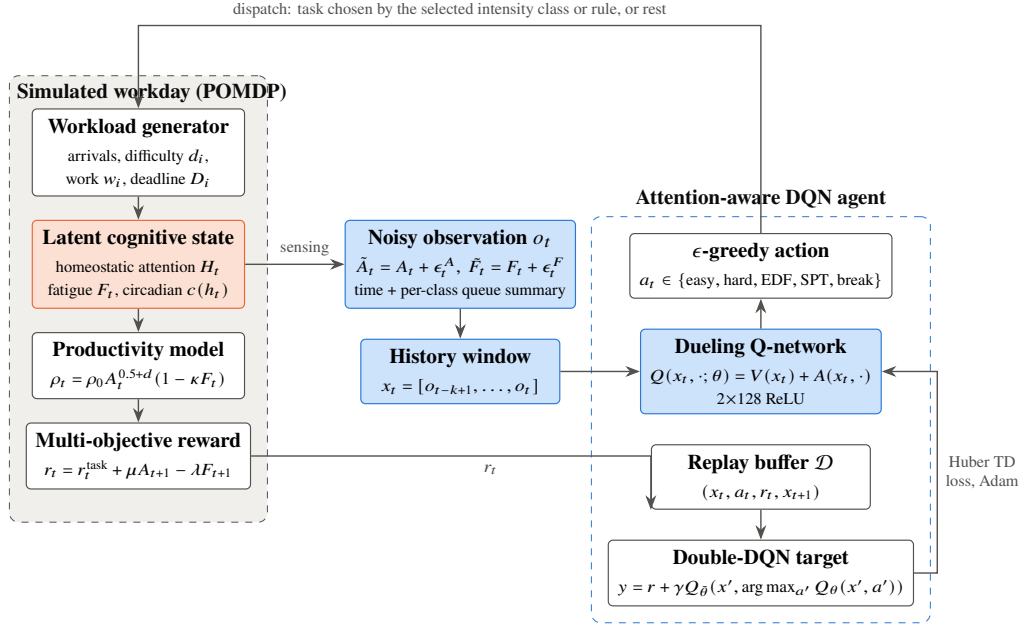


Figure 2: Overview of the attention-aware scheduling framework. The worker’s cognitive state is latent; the agent receives noisy estimates together with a queue summary, encodes a short observation history, and selects an intensity class, a dispatching rule or a break. Transitions are stored in a replay buffer and used to train a dueling Q-network with Double-DQN targets.

283 4.1. From Q-learning to deep Q-networks

284 For a fully observable MDP, the optimal action-value function satisfies the Bell-
 285 man optimality equation $Q^*(s, a) = \mathbb{E}[r + \gamma_{RL} \max_{a'} Q^*(s', a') \mid s, a]$ [137, 138],
 286 and tabular Q-learning converges to Q^* under standard conditions [84]. With con-
 287 tinuous cognitive variables, a tabular representation is infeasible, so Q is approx-
 288 imated by a neural network $Q(\cdot, \cdot; \theta)$ trained to minimise the temporal-difference

289 (TD) error. The DQN of Mnih et al. [85] stabilises this nonlinear approximation
 290 with an experience replay buffer [128], which breaks the temporal correlation
 291 of samples, and a periodically synchronised target network $\bar{\theta}$. Convergence is
 292 not guaranteed for nonlinear function approximation [130], so we assess stability
 293 empirically across independent seeds.

294 4.2. *Handling partial observability with an observation history*

295 In a POMDP, the optimal policy is a function of the belief $b_t(s) = \Pr(s_t =$
 296 $s \mid o_{0:t}, a_{0:t-1})$ [57, 59]. Exact belief tracking is intractable for our continuous,
 297 nonlinear dynamics, and a memoryless policy $\pi(o_t)$ can be arbitrarily suboptimal
 298 [64]. Two structural properties of the scheduling problem make a short history
 299 informative. First, the latent state is persistent. H_t and F_t change slowly relative
 300 to the observation noise, so averaging several consecutive estimates reduces the
 301 error: for k independent observations of a quasi-static quantity, the standard error
 302 falls by a factor $\sqrt{k}$. Second, the trajectory of the observations, for example fatigue
 303 rising after consecutive hard tasks or attention recovering after a break, reveals the
 304 direction and rate of change, which a single noisy reading cannot. We therefore
 305 feed the network the window

$$x_t = [o_{t-k+1}, \dots, o_t] \in \mathbb{R}^{13k}, \quad k = 4, \quad (9)$$

306 zero-padded at the start of the day. This is the finite-memory approximation used
 307 for Atari [85]. It is computationally cheaper than recurrent alternatives [65, 127]
 308 and is a strong baseline for many POMDPs [66]. A 40-minute window spans the
 309 time scale of a typical work bout. Section 6.6 varies $k \in \{1, 2, 4, 8\}$.

310 4.3. Double Q -learning and dueling architecture

311 Max-based targets overestimate action values in noisy environments [67], and
 312 our environment is noisy by design. AA-DQN therefore uses the Double-DQN
 313 target, which decouples action selection (online network) from evaluation (target
 314 network):

$$y_t = r_t + \gamma_{\text{RL}}(1 - \text{done}_t) Q(x_{t+1}, \arg \max_{a'} Q(x_{t+1}, a'; \theta); \bar{\theta}). \quad (10)$$

315 In many states, for example at the start of the day or when the queue is empty, the
 316 choice between easy and hard work matters much less than the value of the state
 317 itself. A dueling head [68] represents this explicitly:

$$Q(x, a; \theta) = V(x; \theta) + \left(A(x, a; \theta) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(x, a'; \theta) \right). \quad (11)$$

318 The shared body consists of two fully connected layers of 128 ReLU units. The
 319 loss is the Huber loss of the TD error, which is robust to the occasional large
 320 reward of a hard task completed on time [139]:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,a,r,x') \sim \mathcal{D}} [\ell_{\delta=1}(Q(x, a; \theta) - y)]. \quad (12)$$

321 Parameters are updated with Adam [140] and gradient-norm clipping. The vanilla
 322 DQN baseline uses the same action set and observation but $k = 1$, the standard max
 323 target and a single linear output head. This isolates the contribution of memory
 324 and of the two stabilisation components.

325 4.4. Exploration and training procedure

326 Actions are selected ϵ -greedily with an exponentially decaying exploration
 327 rate $\epsilon_n = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})e^{-\kappa_{\epsilon} n}$, where n is the environment step, $\epsilon_{\max} = 1$
 328 and $\epsilon_{\min} = 0.01$, and κ_{ϵ} is set so that ϵ reaches 0.05 after half of the training

Algorithm 1 Training of the attention-aware DQN (AA-DQN).

Require: episodes E , history length k , replay capacity $|\mathcal{D}|$, batch size B , target period C , validation days $\mathcal{V}$

- 1: initialise θ randomly, $\bar{\theta} \leftarrow \theta$, $\mathcal{D} \leftarrow \emptyset$, $n \leftarrow 0$, $J^* \leftarrow -\infty$
- 2: **for** episode $e = 1, \dots, E$ **do**
- 3: sample a workday (tasks, initial state, noise); observe o_0 ; $x_0 \leftarrow [\mathbf{0}, \dots, \mathbf{0}, o_0]$
- 4: **for** $t = 0, \dots, T - 1$ **do**
- 5: $a_t \leftarrow$ random action with prob. ϵ_n , else $\arg \max_a Q(x_t, a; \theta)$
- 6: dispatch a_t (EDF within class or rest); simulate Eqs. (1)–(4); receive r_t (Eq. (8))
and o_{t+1}
- 7: $x_{t+1} \leftarrow \text{shift}(x_t, o_{t+1})$; store $(x_t, a_t, r_t, x_{t+1}, \text{done}_t)$ in $\mathcal{D}$; $n \leftarrow n + 1$
- 8: **if** $|\mathcal{D}| \geq \text{warm-up}$ **then**
- 9: sample a minibatch of B transitions; compute targets with Eq. (10)
- 10: $\theta \leftarrow \theta - \eta_{lr} \text{Adam}(\nabla_{\theta} \mathcal{L}(\theta))$ with gradient clipping
- 11: **if** $n \bmod C = 0$ **then** $\bar{\theta} \leftarrow \theta$
- 12: **if** $e \bmod 100 = 0$ **then**
- 13: $J \leftarrow$ mean greedy return on $\mathcal{V}$; **if** $J > J^*$ **then** $J^* \leftarrow J$, $\theta^* \leftarrow \theta$
- 14: **return** θ^*

329 budget. Algorithm 1 summarises training. Every 100 episodes, the greedy policy
330 is evaluated on a fixed set of 100 *validation* days that are disjoint from the training
331 and test days, and the parameters with the highest validation return are retained.
332 This mirrors the tuning of the heuristic baselines on their own held-out days
333 (Section 5.1), so that both families receive the same model-selection budget and
334 neither sees the test days. For transparency we also report the final-iterate policy
335 (Supplementary Table S2).

336 4.5. Computational complexity

337 A forward pass costs $O(13k \cdot 128 + 128^2 + 128 \cdot |\mathcal{A}|) \approx 2.4 \times 10^4$ multiply–
338 accumulate operations, independent of the number of tasks, because the observa–
339 tion summarises the queue per class. The dispatch step is $O(N \log N)$. Inference
340 therefore takes microseconds on a CPU (Section 6.9), which is negligible rela–
341 tive to a 10-minute decision interval and allows deployment on wearable or edge
342 devices.

343 5. Experimental design

344 5.1. Baselines

345 We compare AA-DQN against the vanilla DQN and nine rule-based sched–
346 ulers, which span classical dispatching and hand-crafted cognition-aware policies.
347 Baselines dispatch individual tasks directly, so they have at least as much control
348 over task choice as the RL agents.

- 349 • **Random:** uniform choice among EASY, HARD and BREAK.
- 350 • **FIFO:** the available task that arrived first; never rests.
- 351 • **EDF:** the available task with the earliest deadline; never rests [2, 5].
- 352 • **SPT:** the available task with the least remaining work; never rests [1].
- 353 • **EDF-Periodic:** EDF with one break after every p work slots, a fixed work–
354 rest cycle in the spirit of ergonomic rest-allowance guidance [34].
- 355 • **EDF-Threshold:** EDF that rests whenever the *observed* fatigue exceeds τ_F
356 or the observed attention falls below τ_A . This is a reactive, state-aware rule.

- **SPT-Periodic** and **SPT-Threshold**: the same two rest rules combined with SPT sequencing. SPT limits the damage of overload by finishing short tasks first, so these variants combine the best classical sequencing rule (Section 6.2) with state-aware rest. They were added after a pre-submission review identified them as the strongest hand-designed competitors (Supplementary Section S6).
- **Cog-Heuristic**: rests on the same threshold rule and otherwise schedules a hard task only when observed attention is at least τ_H , an easy task otherwise (EDF within class). This hand-crafted rule encodes the intuition that the learned policy might discover.

The parameters of the five parametric heuristics were selected by exhaustive grid search, maximising mean return on 300 *tuning* days that are disjoint from the training, validation and test days. The grids were $p \in \{2, 3, 4, 5, 6, 8, 10\}$; $\tau_F \in \{0.3, \dots, 0.9, 1.01\}$ and $\tau_A \in \{0, 0.2, \dots, 0.7\}$ (56 combinations, for both threshold rules); and, for Cog-Heuristic, $\tau_F \times \tau_A \times \tau_H$ with $\tau_H \in \{0.4, \dots, 0.9\}$ (252 combinations). The heuristics observe exactly the same noisy signals as the agents. The selected parameters are reported in Supplementary Table S1. Tuning is repeated separately for every environment variant, so that each ablation compares the agent with the best heuristic *for that variant*.

5.2. Protocol

The main configurations were trained with ten independent seeds, the environment ablations and the domain-randomised variant with five, and the reward-weight sweep and the component study with three, each for 5000 episodes (2.4×10^5 decisions). Four disjoint sets of days were used: training days (generated from the

381 training seed), 100 validation days for checkpoint selection, 300 tuning days for
382 heuristic parameters, and 500 test days used only once, for the final evaluation.
383 All methods are evaluated greedily on the *same* 500 test days with *common ran-*
384 *dom numbers*. Each test day fixes the task set, the initial cognitive state and the
385 sequences of process and observation noise, so differences between methods are
386 paired and not confounded by sampling variation. Seven further seeds of the
387 intensity-only variant were trained for the secondary analysis (Section 6.7). In
388 total, 82 agents were trained and more than 10^5 test days were simulated.

389 5.3. Metrics

390 Besides the return (Eq. (8)), which depends on the reward weights, we re-
391 port reward-independent operational metrics: the *on-time rate* (fraction of the
392 day’s tasks completed by their deadline); the *difficulty-weighted on-time rate*
393 $\sum_i d_i \mathbb{I}(\text{on time}) / \sum_i d_i$; the *completion rate* (completed at any time); the mean
394 *true* attention $\bar{A}$ and fatigue $\bar{F}$ over the day; the *high-fatigue exposure*, i.e. the
395 fraction of slots with $F_t > 0.7$; and the *break share*.

396 5.4. Statistical analysis

397 Following recommendations for reliable RL evaluation [60, 61], we treat the
398 training seed as the unit of replication for learned policies. For each seed, metrics
399 are averaged over the 500 test days. We report the mean and standard deviation
400 across seeds, and 95% bootstrap confidence intervals [141] with 5000 resamples.
401 Deterministic heuristics are summarised across test days. To compare AA-DQN
402 with each baseline, we average AA-DQN over seeds for each test day and apply the
403 two-sided Wilcoxon signed-rank test [142] to the 500 paired day-level differences.
404 We report the matched-pairs rank-biserial correlation r_{rb} as effect size, the win

405 rate, and Holm-adjusted p -values [143] over the nine baseline comparisons. As
406 a more conservative seed-level check, we also report how many of the ten seeds
407 outperform each baseline.

408 5.5. *Implementation*

409 The simulator, agents and baselines are implemented in Python with NumPy
410 and PyTorch [144]. Table 4 lists the hyperparameters. Training used one CPU
411 thread per run. The code, configuration files, raw per-day results and analysis
412 scripts are released (see Data availability) so that every number and figure in this
413 paper can be regenerated with a single command.

Table 4: Agent hyperparameters (shared by DQN and AA-DQN unless stated).

Hyperparameter	DQN	AA-DQN
History length k	1	4
Target	max (standard)	Double DQN
Output head	linear	dueling ($V + \text{advantage}$)
Hidden layers		2×128 , ReLU
Optimiser, learning rate		Adam, 5×10^{-4}
Discount γ_{RL}		0.99
Replay capacity, warm-up		50 000, 1 000 transitions
Minibatch size		64
Target update period C		1 000 steps (hard copy)
Loss, gradient clip		Huber ($\delta = 1$), $\ \nabla\ _2 \leq 10$
Exploration	ϵ : $1.0 \rightarrow 0.01$, exponential (0.05 at 50% of budget)	
Action set	{EASY, HARD, EDF, SPT, BREAK}	
Training budget	5 000 episodes = 240 000 decisions	
Checkpoint selection	best greedy return on 100 validation days, every 100 episodes	

414 6. Results

415 6.1. Learning dynamics

416 Figure 3 shows greedy validation performance during training, averaged over
417 ten seeds. AA-DQN improves rapidly during the first 1000 episodes and slowly
418 thereafter; it is still improving at the end of the 5000-episode budget. The seed-
419 averaged curve of the *current* parameters remains below the best tuned heuristic
420 (SPT-Threshold) throughout training, in both return and on-time rate. This average,
421 however, hides substantial variation between checkpoints within each run: the

best checkpoints of every seed reach the heuristic’s level, while neighbouring checkpoints fall below it. Such policy oscillation is typical of value-based deep RL with a constant learning rate [60], and it motivates selecting checkpoints on held-out validation days. Validation-selected checkpoints were taken at episode 3350 on average for AA-DQN and 3490 for DQN. Because validation and test days are disjoint, the test results below are not inflated by this selection. The vanilla DQN learns more slowly and plateaus well below AA-DQN. Its training return is also less stable (Supplementary Fig. S2), consistent with the over-estimation that Double Q-learning mitigates [67] and with the known weakness of memoryless policies in POMDPs [64].

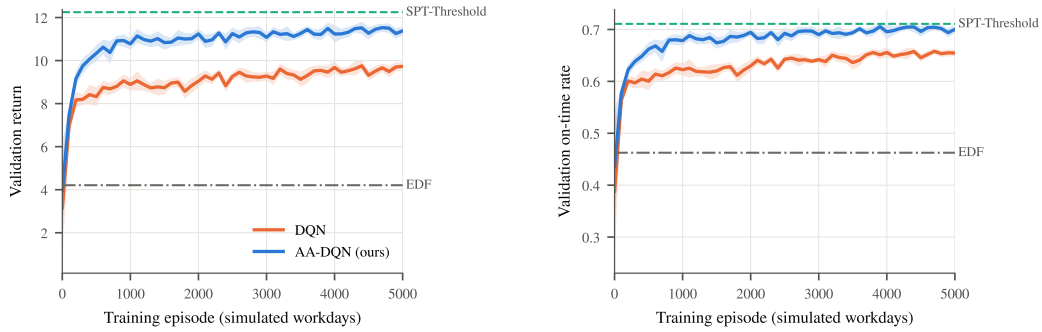


Figure 3: Greedy validation performance of the current parameters during training (mean of ten seeds; shaded bands are 95% confidence intervals). Horizontal lines show the best tuned heuristic (SPT-Threshold) and EDF on the same 100 validation days. Test results use the validation-selected checkpoint of each seed.

6.2. Comparison with baselines

Table 5 and Figs. 4–5 report performance on the 500 held-out test days, and Table 6 gives the paired statistical comparisons for the pre-specified primary agent. The intensity-only variant shown in the same table and figures is a secondary

436 analysis, discussed in Section 6.7. Four findings stand out.

437 (i) *Capacity-agnostic rules fail under cognitive dynamics.* The rules that never
438 rest (FIFO, EDF, SPT) drive the worker into a low-capacity regime: mean fatigue
439 reaches 0.48–0.55, attention falls to 0.46–0.54, and EDF completes only 46.8% of
440 tasks on time. EDF performs particularly poorly, which reproduces the overload
441 domino effect known from real-time scheduling [70]: once capacity falls, chasing
442 the most urgent task means working on tasks that will be late anyway. SPT limits
443 the damage by finishing short tasks first (62.5%).

444 (ii) *State-aware rest is the dominant lever.* Adding rest to these rules, either
445 periodically or when observed fatigue or attention crosses a tuned threshold, raises
446 the return by 4–8 units and roughly halves mean fatigue. All competitive policies
447 rest in 15–20% of slots. The best rule overall is SPT-Threshold (return 12.30,
448 on-time rate 71.4%), which combines SPT sequencing with threshold-triggered
449 rest.

450 (iii) *AA-DQN matches the best tuned rule on return and is the only policy that is*
451 *strong on both completion measures.* AA-DQN achieves a return of 12.17 ± 0.15
452 across seeds. This is statistically indistinguishable from SPT-Threshold ($\Delta =$
453 -0.13 , 95% CI $[-0.29, 0.02]$, $p_{\text{Holm}} = 0.24$), and it significantly exceeds every
454 other baseline ($p_{\text{Holm}} < 10^{-4}$). It attains the highest on-time rate of all methods
455 (72.8%), significantly above SPT-Threshold (+1.4 percentage points $[0.7, 2.1]$,
456 $p_{\text{Holm}} = 0.00016$, all ten seeds better). The difficulty-weighted on-time rate reveals
457 why the two leading families differ (Fig. 5). SPT-based rules complete many tasks
458 on time by finishing short tasks first, but they neglect hard tasks (weighted on-time
459 rate 59.7%). EDF-based rules protect hard tasks (65.7%) but complete fewer tasks
460 overall. AA-DQN combines the strengths of both: it completes as many tasks on

time as any rule, while matching the EDF rules on difficulty-weighted completion (65.5%; +5.7 pp over SPT-Threshold, [4.9, 6.6]). In other words, the learned policy avoids the starvation of long tasks that is the classical weakness of SPT [4]. No single hand-designed rule in our set achieves this.

(iv) *The learned policy pays for this with somewhat more fatigue and fewer late completions.* AA-DQN rests less than the tuned heuristics (15.0% versus 18.7% of slots for SPT-Threshold). Its mean fatigue is therefore higher (0.333 versus 0.268), although still 39% below EDF, and fatigue never exceeds 0.7. Because it focuses effort on tasks that can still meet their deadline, it also completes fewer tasks eventually (77.1% versus 81.7% for SPT-Threshold, counting late completions). This explains why its return, which rewards late completions and low fatigue, is not higher despite the higher on-time rate. Section 6.5 shows that the fatigue level can be shifted by the reward weight.

Relative to the memoryless vanilla DQN, adding memory, Double Q-learning and the dueling head increases the return by 1.88 ([1.81, 1.96]), with a win on 98.0% of test days. The vanilla DQN would rank only seventh among the eleven methods, which shows that a naive application of DQN to this POMDP would wrongly suggest that learning is not competitive. Validation-based checkpoint selection matters for both agents (Supplementary Table S2): the final-iterate AA-DQN has a return of 11.55 and an on-time rate of 71.0%, and the final-iterate DQN has a return of 9.84. We report the validation-selected policies as primary results because the heuristics received an equivalent selection step (grid search) on their own held-out days.

Table 5: Performance on 500 held-out test days (nominal environment). Learned policies: mean $\pm$ s.d. across ten training seeds of the per-seed test means. Heuristics: mean $\pm$ s.d. across test days (they are deterministic given the day). Bold marks the best value in each column.

Method	Return	On-time rate	Weighted on-time	Mean attention	Mean fatigue	Break share
Random	8.85 ± 2.71	0.535 ± 0.131	0.488 ± 0.143	0.855 ± 0.037	0.207 ± 0.045	0.326 ± 0.065
FIFO	3.49 ± 3.49	0.432 ± 0.139	0.422 ± 0.125	0.463 ± 0.106	0.546 ± 0.045	0.000 ± 0.003
EDF	4.24 ± 3.82	0.468 ± 0.155	0.454 ± 0.137	0.465 ± 0.105	0.545 ± 0.046	0.000 ± 0.003
SPT	7.95 ± 2.94	0.625 ± 0.120	0.494 ± 0.122	0.543 ± 0.093	0.477 ± 0.038	0.000 ± 0.001
EDF-Periodic	11.41 ± 3.32	0.660 ± 0.160	0.657 ± 0.148	0.798 ± 0.040	0.296 ± 0.029	0.188 ± 0.005
EDF-Threshold	11.35 ± 3.28	0.654 ± 0.164	0.650 ± 0.155	0.806 ± 0.022	0.288 ± 0.034	0.201 ± 0.037
SPT-Periodic	12.25 ± 2.04	0.709 ± 0.096	0.591 ± 0.111	0.817 ± 0.034	0.261 ± 0.023	0.188 ± 0.005
SPT-Threshold	12.30 ± 2.03	0.714 ± 0.099	0.597 ± 0.114	0.813 ± 0.022	0.268 ± 0.036	0.187 ± 0.035
Cog-Heuristic	10.59 ± 2.76	0.638 ± 0.132	0.533 ± 0.150	0.803 ± 0.034	0.279 ± 0.020	0.198 ± 0.025
DQN	10.29 ± 0.26	0.677 ± 0.008	0.579 ± 0.012	0.677 ± 0.017	0.394 ± 0.018	0.108 ± 0.020
AA-DQN (ours, primary)	12.17 ± 0.15	0.728 ± 0.005	0.655 ± 0.012	0.754 ± 0.011	0.333 ± 0.012	0.150 ± 0.018
AA-DQN, intensity actions (secondary)	12.61 ± 0.18	0.740 ± 0.005	0.678 ± 0.008	0.770 ± 0.012	0.319 ± 0.014	0.162 ± 0.016

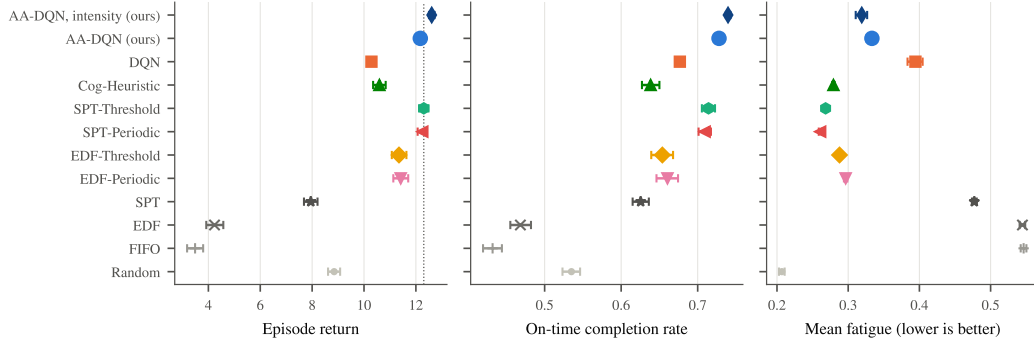


Figure 4: Test performance with 95% bootstrap confidence intervals (learned policies: across seeds; heuristics: across days). The dotted line marks the best heuristic return.

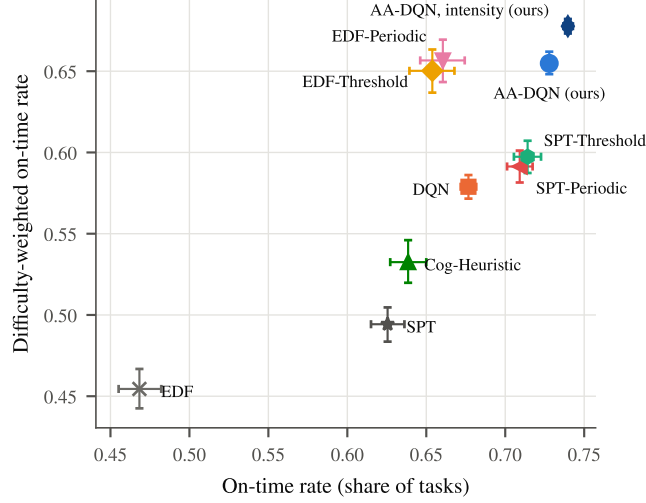


Figure 5: Count-based versus difficulty-weighted on-time completion (95% CIs). SPT-based rules complete many tasks but neglect hard ones; EDF-based rules protect hard tasks but complete fewer. The two AA-DQN variants are the only policies near the best value on both axes.

Table 6: Paired comparison of AA-DQN with each baseline on 500 common-random-number test days: mean difference with 95% bootstrap CI (on-time rates in percentage points), matched-pairs rank-biserial correlation r_{rb} (positive favours AA-DQN), and Holm-adjusted Wilcoxon signed-rank p -values.

AA-DQN vs.	Episode return			On-time rate (pp)			Weighted on-time rate (pp)		
	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}
Random	+3.32 [+3.11, +3.52]	0.97	$< 10^{-4}$	+19.3 [+18.3, +20.3]	1.00	$< 10^{-4}$	+16.7 [+15.7, +17.7]	0.98	$< 10^{-4}$
FIFO	+8.68 [+8.46, +8.90]	1.00	$< 10^{-4}$	+29.6 [+28.7, +30.5]	1.00	$< 10^{-4}$	+23.3 [+22.5, +24.0]	1.00	$< 10^{-4}$
EDF	+7.93 [+7.70, +8.17]	1.00	$< 10^{-4}$	+26.0 [+25.0, +26.9]	1.00	$< 10^{-4}$	+20.0 [+19.3, +20.7]	1.00	$< 10^{-4}$
SPT	+4.22 [+4.05, +4.39]	1.00	$< 10^{-4}$	+10.2 [+9.6, +10.9]	0.96	$< 10^{-4}$	+16.1 [+15.3, +16.8]	1.00	$< 10^{-4}$
EDF-Periodic	+0.76 [+0.57, +0.94]	0.34	$< 10^{-4}$	+6.7 [+5.9, +7.6]	0.69	$< 10^{-4}$	-0.2 [-0.9, +0.5]	-0.05	0.6403
EDF-Threshold	+0.82 [+0.64, +1.01]	0.36	$< 10^{-4}$	+7.4 [+6.6, +8.3]	0.72	$< 10^{-4}$	+0.5 [-0.2, +1.2]	0.01	0.8269
SPT-Periodic	-0.08 [-0.24, +0.08]	-0.06	0.2860	+1.9 [+1.1, +2.6]	0.24	$< 10^{-4}$	+6.3 [+5.5, +7.2]	0.62	$< 10^{-4}$
SPT-Threshold	-0.13 [-0.29, +0.02]	-0.08	0.2390	+1.4 [+0.7, +2.1]	0.20	0.0002	+5.7 [+4.9, +6.6]	0.61	$< 10^{-4}$
Cog-Heuristic	+1.58 [+1.43, +1.73]	0.80	$< 10^{-4}$	+8.9 [+8.3, +9.6]	0.92	$< 10^{-4}$	+12.2 [+11.4, +13.1]	0.93	$< 10^{-4}$

484 6.3. What does the agent learn?

485 Figure 6 shows how actions and latent states evolve through the day, aggregated
486 over the 500 test days (and over the ten seeds for AA-DQN). Three behavioural
487 regularities distinguish the learned policy from the heuristics.

488 *Work while fresh.* AA-DQN almost never rests in the first hour (0.5% of slots,
489 versus 3.6% for SPT-Threshold, 4.6% for EDF-Threshold and a fixed 16.7% for
490 EDF-Periodic). It exploits the high morning attention and lets fatigue rise to ≈ 0.38
491 before it starts to insert breaks. A fixed cycle cannot distinguish a fresh worker
492 at 09:30 from a recovered one at 15:00, and a threshold rule reacts only once a
493 threshold has been crossed.

494 *Rest when work is least productive.* The break share of AA-DQN rises to a peak
495 of 24.6% between 13:00 and 14:00, which coincides with the modelled post-lunch
496 dip, and declines afterwards. In the 13:00–15:00 window it rests in 22.4% of slots,
497 versus 12.5% at other times. Resting is cheapest when the productivity it forgoes
498 is lowest (Eq. (4)). The threshold rules show a similar but sharper peak (35–36%
499 between 13:00 and 14:00) only because the dip pushes observed attention below
500 their fixed threshold.

501 *Match difficulty to capacity without starving hard tasks.* AA-DQN devotes
502 about half of the morning slots to hard tasks (50% at 09:00–11:00), reduces this to
503 one third around the afternoon trough (33% at 13:00–15:00), and returns to hard
504 work in the last hour (48%) as their deadlines approach. SPT-Threshold behaves
505 very differently. It does almost no hard work in the first hour (8%) because hard
506 tasks are long, postpones them until they become urgent, and then spends 82% of
507 the last hour on hard work, when attention is lowest and hard tasks are slowest
508 (Eq. (4)). This back-loading explains its low difficulty-weighted on-time rate

509 (Section 6.2). EDF, in contrast, does more hard work in the afternoon (71%) than in
 510 the morning (60%) and never rests. The hand-crafted Cog-Heuristic was designed
 511 to capture the difficulty–attention intuition but, with a single attention threshold,
 512 also pushes hard tasks into the late afternoon (67% after 16:00). The learned policy
 513 is the only one that front-loads demanding work while still finishing short tasks,
 514 which illustrates how hard it is to encode such time-, state- and deadline-dependent
 515 trade-offs in rules.

516 Compared with the memoryless DQN, AA-DQN rests more (15.0% versus
 517 10.8% of slots) and takes breaks at a lower true fatigue level (0.36 versus 0.44).
 518 The DQN concentrates its rest in the last hour (29% of slots after 16:00), consistent
 519 with late, reactive breaks triggered by single noisy readings.

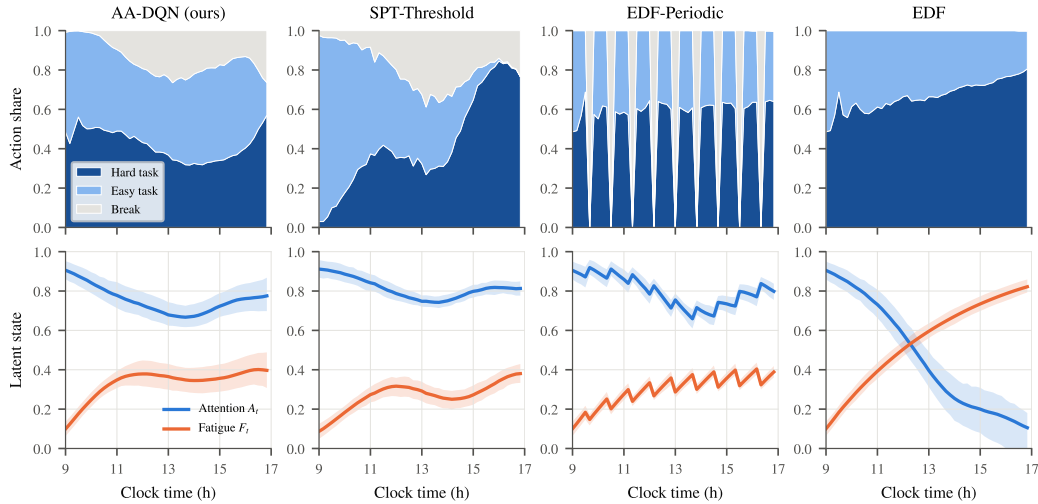


Figure 6: Learned and heuristic behaviour across the workday on 500 test days. Top: share of slots spent on hard tasks, easy tasks and breaks. Bottom: mean true attention and fatigue (bands: interquartile range). The saw-tooth pattern of EDF-Periodic reflects its fixed work–rest cycle.

520 *6.4. Environment ablations: when does learning add value?*

521 To separate the contributions of the environment’s components, we retrained
522 AA-DQN (five seeds) in four variants and re-tuned every heuristic for each variant.
523 Table 7 and Fig. 7 report the difference between AA-DQN and the best heuristic
524 *within* each variant. This avoids comparing returns across variants with different
525 reward scales.

526 *Observation quality is the main bottleneck.* When attention and fatigue are
527 observed without noise, AA-DQN improves to 13.61 and exceeds the best heuristic
528 (SPT-Periodic, 12.25) by +1.36 ([1.20, 1.52]) in return and +5.5 percentage points
529 in on-time rate. Its mean fatigue also falls, from 0.333 in the nominal setting
530 to 0.269. The heuristics do not benefit from perfect information at all: the best
531 threshold rule is no better than with noisy observations, because a fixed threshold
532 cannot use precise state information beyond the threshold crossing. Learned
533 policies thus have a structural advantage that grows with sensor quality, while the
534 advantage of hand-crafted rules does not.

535 *The circadian process is not what the agent exploits relative to rules.* Without
536 the circadian term, AA-DQN and the best rule are tied in return (+0.06, [-0.12,
537 0.24]), and AA-DQN keeps a +2.5 pp on-time advantage. This is the same pattern
538 as in the full model, so the agent’s behaviour around the post-lunch dip (Section 6.3)
539 is an adaptation to the dynamics rather than the source of an advantage over the
540 threshold rules.

541 *The cognitive reward terms are not responsible for the result.* Trained with
542 a task-only reward ($\lambda = \mu = 0$), AA-DQN still matches the best rule in return
543 (-0.13, [-0.29, 0.04]) and still rests in 11.5% of slots (Table 8), because rest pays
544 off through productivity alone. The agent’s on-time advantage disappears in this

variant (-0.3 pp), which indicates that the attention term shapes *how* it trades completions against cognitive load.

Learning also helps without cognitive dynamics. In the static-capacity variant, in which rest has no value, AA-DQN beats the best rule (SPT) by $+0.91$ ($[0.74, 1.07]$) and $+3.7$ pp. Under overload, combining urgency (EDF) with processing time (SPT) depending on the state of the queue is better than either rule alone, a well-known property of composite dispatching [43, 72]. This result does *not* support the claim, made in an earlier version of this work, that reinforcement learning helps only when cognition is modelled. What cognitive dynamics change is the size of the stakes: in the full model, state-aware policies exceed the capacity-agnostic rules by 4–8 return units (Table 5), versus 1–3 units in the static model.

Table 7: Environment ablations. AA-DQN (five seeds) versus the best re-tuned heuristic within each variant; differences are paired over 500 test days with 95% bootstrap CIs (on-time in percentage points).

Environment variant	AA-DQN return	Best heuristic (return)	Δ return [95% CI]	AA-DQN on-time	Best heuristic (on-time)	Δ on-time [95% CI]
Full model	12.17 ± 0.15	SPT-Threshold (12.30)	$-0.13 [-0.29, +0.02]$	0.728	SPT-Threshold (0.714)	$+0.014 [+0.007, +0.021]$
No observation noise	13.61 ± 0.08	SPT-Periodic (12.25)	$+1.36 [+1.20, +1.52]$	0.764	SPT-Periodic (0.709)	$+0.055 [+0.047, +0.062]$
No circadian process	12.99 ± 0.21	SPT-Periodic (12.93)	$+0.06 [-0.12, +0.24]$	0.754	SPT-Threshold (0.729)	$+0.025 [+0.017, +0.032]$
Task-only reward	11.53 ± 0.18	SPT-Threshold (11.66)	$-0.13 [-0.29, +0.04]$	0.711	SPT-Threshold (0.715)	$-0.003 [-0.011, +0.003]$
Static capacity (no cognition)	10.41 ± 0.10	SPT (9.50)	$+0.91 [+0.74, +1.07]$	0.670	SPT (0.633)	$+0.037 [+0.029, +0.045]$

6.5. Productivity–fatigue trade-off

The fatigue weight λ (with $\mu = \lambda/2$) controls how much the agent values the worker’s state. Figure 8 and Table 8 show the policies obtained for $\lambda \in \{0, 0.05, 0.1, 0.2, 0.4\}$. Increasing λ lowers mean fatigue monotonically, from 0.365 to 0.225, by raising the break share from 11.5% to 30.4%. Over most of this range it does *not* reduce productivity. The on-time rate rises from 70.8%

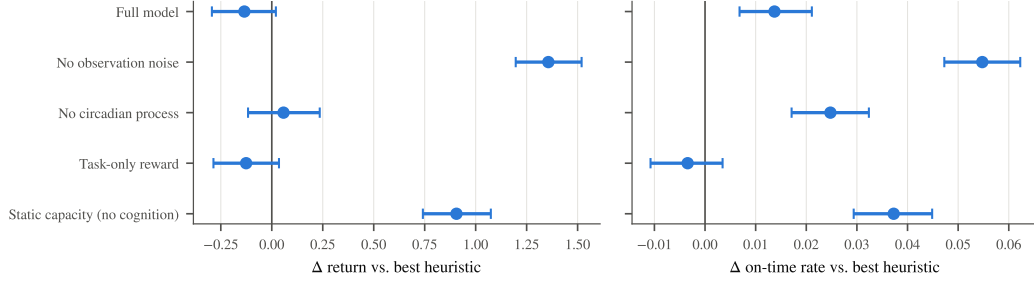


Figure 7: Difference between AA-DQN and the best heuristic within each environment variant (95% CIs). Positive values favour AA-DQN.

562 ($\lambda = 0$) to 73.1% ($\lambda = 0.2$), and is still 72.3% at $\lambda = 0.4$. At $\lambda = 0.4$, AA-DQN
 563 keeps the worker *less* fatigued than any of the tuned heuristics (Fig. 8) while
 564 completing more tasks on time than the best of them (71.4%). Moderate concern
 565 for the worker’s state is therefore not a cost to productivity but a means to it, which
 566 echoes empirical findings on rest breaks [18, 19, 22]. The fatigue penalty of the
 567 nominal agent noted in Section 6.2 is a consequence of the chosen weight, not of
 568 the method.

Table 8: Effect of the reward weight λ on AA-DQN (mean $\pm$ s.d. over three seeds; 500 test days).

λ ($\mu = \lambda/2$)	On-time rate	Mean fatigue	High-fatigue exposure	Break share
0	0.708 ± 0.009	0.365 ± 0.015	0.000	0.115
0.05	0.718 ± 0.009	0.354 ± 0.009	0.000	0.124
0.1	0.727 ± 0.005	0.344 ± 0.002	0.000	0.137
0.2	0.731 ± 0.007	0.304 ± 0.010	0.000	0.181
0.4	0.723 ± 0.012	0.225 ± 0.008	0.000	0.304

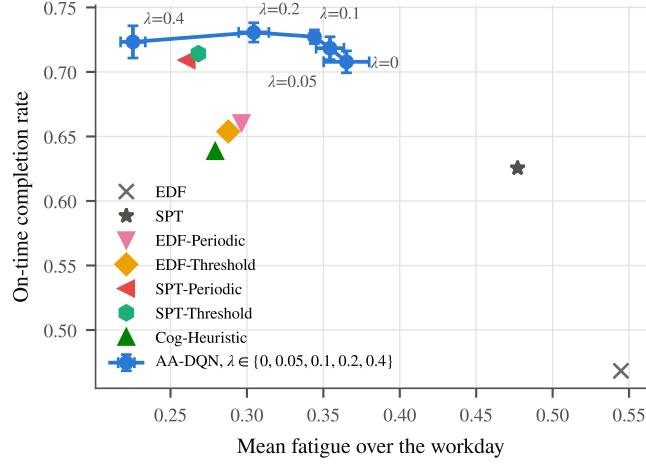


Figure 8: Productivity–fatigue trade-off. AA-DQN trained with different fatigue weights λ (three seeds each; error bars: s.d.) versus the tuned heuristics.

569 6.6. Component analysis

570 Table 9 isolates the components of AA-DQN (three seeds each, except for the
571 ten-seed configurations). *Memory is essential.* Removing the observation history
572 ($k = 1$) reduces the return from 12.17 to 10.34, close to the vanilla DQN, and
573 a two-step history recovers only part of the loss (11.17). An eight-step history
574 performs at least as well as $k = 4$ (12.48). A GRU encoder over an eight-step
575 window performs similarly (12.54), so the benefit comes from the length of the
576 memory rather than from recurrence per se. In contrast, Double Q-learning and the
577 dueling head have small effects within seed variability (12.07 and 12.28 without
578 them). This indicates that the partial observability of the cognitive state, not value
579 over-estimation, is the main difficulty of the problem.

580 *A smaller action space learns better.* The agent restricted to the intensity ac-
581 tions {EASY, HARD, BREAK} achieved the highest return of all configurations in the
582 three-seed study (12.61). Because this variant was not our pre-specified primary

method, we did not promote it on the basis of three seeds. Instead, we re-evaluated it with the full ten-seed protocol as a secondary analysis (Section 6.7). The result is plausible in hindsight. With the shortest-task feature in the observation, choosing an easy (typically short) task already captures much of the benefit of SPT, while the larger hybrid action space slows exploration and credit assignment.

Table 9: Component and sensitivity analysis (mean $\pm$ s.d. over seeds; 500 test days).

Configuration	Seeds	Return	On-time rate	Train time (s)
AA-DQN (hybrid actions, $k=4$, Double, Dueling)	10	12.17 ± 0.15	0.728 ± 0.005	720
– without history ($k=1$)	3	10.34 ± 0.19	0.683 ± 0.010	551
– without Double Q-learning	3	12.07 ± 0.19	0.720 ± 0.014	552
– without dueling head	3	12.28 ± 0.04	0.725 ± 0.004	517
– intensity-only actions {easy, hard, break}	10	12.61 ± 0.18	0.740 ± 0.005	582
history $k=2$	3	11.17 ± 0.12	0.701 ± 0.001	568
history $k=8$	3	12.48 ± 0.37	0.733 ± 0.011	605
GRU encoder over $k=8$ history	3	12.54 ± 0.16	0.728 ± 0.006	2463
Vanilla DQN ($k=1$, no Double/Dueling)	10	10.29 ± 0.26	0.677 ± 0.008	614

6.7. Secondary analysis: intensity-only agent

The component study (Section 6.6) suggested that restricting AA-DQN to the intensity actions {EASY, HARD, BREAK} improves performance. We therefore trained seven additional seeds of this variant and evaluated it with the same ten-seed protocol as the primary agent. Because the variant was selected after a look at test-set results from three seeds, we report it as a secondary, hypothesis-generating analysis and keep the pre-specified hybrid agent as the primary result.

With ten seeds, the intensity-only agent achieves a return of 12.61 ± 0.18 and an on-time rate of 74.0%. Relative to the best rule, SPT-Threshold, the paired

597 differences are 0.31 in return ($[0.13, 0.48]$, $p_{\text{Holm}} = 0.00016$), 2.5 pp in on-time
 598 rate ($[1.8, 3.3]$, $p_{\text{Holm}} < 10^{-4}$) and 8.0 pp in difficulty-weighted on-time rate ($[7.2,$
 599 $8.9]$). Relative to the primary hybrid agent, the differences are 0.44 in return
 600 ($[0.38, 0.50]$) and 1.2 pp in on-time rate. The mean fatigue is 0.319 and the
 601 break share 16.2%. Taken together, the primary and secondary analyses show
 602 that a memory-augmented DQN reaches, and in its compact variant exceeds, the
 603 performance of the best of nine tuned rules. They also show that a compact,
 604 cognitively meaningful action space is preferable to a larger rule-selection space
 605 for this problem.

606 6.8. Robustness and generalisation

607 Finally, we evaluated all policies *zero-shot*, without retraining or re-tuning,
 608 under shifted conditions: different levels of sensor noise at test time, resilient and
 609 fatigue-prone worker profiles, and lighter or heavier workloads (Fig. 9; full results
 610 in Supplementary Table S4). We also trained five seeds of a domain-randomised
 611 agent (AA-DQN-DR) on randomly sampled workloads ($N \in \{8, \dots, 16\}$), worker
 612 profiles ($\pm 30\%$ on all rate parameters) and noise levels ($\sigma_o \in [0.05, 0.2]$).

613 *Sensor noise.* Both AA-DQN variants are insensitive to the test-time noise
 614 level. Their returns vary by less than 0.15 between $\sigma_o = 0$ and $\sigma_o = 0.3$ (pri-
 615 mary 12.14–12.27, intensity-only 12.57–12.71). The threshold rules, in contrast,
 616 degrade as noise grows, because spurious threshold crossings trigger unneces-
 617 sary breaks. SPT-Threshold falls from 12.30 to 11.35 and from 71.4% to 65.6%
 618 on-time, and EDF-Threshold falls from 11.35 to 9.63. The periodic rules are
 619 unaffected, because they ignore the cognitive signals altogether. Learned policies
 620 therefore degrade more gracefully than reactive rules when state estimation is poor.
 621 Their history-based estimate averages out noise that a threshold applied to a single

622 reading cannot.

623 *Worker profiles.* The ranking depends on the profile. For fatigue-prone work-
624 ers, the learned policies (primary 10.05, intensity-only 10.40) and the SPT-based
625 rules (10.21–10.59) form the leading group, far ahead of the EDF-based rules
626 (7.5–7.9). For resilient workers, who need less rest, the EDF-based rules attain the
627 highest return (EDF-Periodic 15.01), with the intensity-only agent close behind
628 (14.76) and achieving the highest on-time rate (81.1%). No single policy is best
629 for every profile, which argues for personalisation (Section 7.5).

630 *Workload.* Generalisation to different numbers of tasks per day is the clearest
631 weakness of the learned policies. With 8 tasks per day the threshold rules com-
632 plete almost all tasks on time and outperform the agents (EDF-Threshold 13.12
633 versus 11.71–12.20). With 16 tasks the SPT-based rules are best (SPT-Threshold
634 11.61 versus 9.69–10.03). Although the observation is normalised by N , the queue
635 statistics of these days lie outside the training distribution. Domain randomisation
636 improved the 16-task result (10.35) and kept on-time rates close to the nominally
637 trained agent across conditions, but it did not close the gap and lowered nominal
638 performance (11.63). A simple randomisation over conditions with checkpoint
639 selection on nominal days is therefore insufficient. Workload-aware training cur-
640 ricula or explicit conditioning on the workload level are needed for deployment
641 across heterogeneous workloads.

642 6.9. Computational cost

643 Training one AA-DQN agent for 5000 simulated workdays took 720 s on aver-
644 age on a single CPU thread (vanilla DQN: 614 s), including the periodic validation
645 runs. The complete evaluation loop, including the Python simulator, takes about
646 97 μ s per decision when 500 days are evaluated in a batch. Even unbatched,

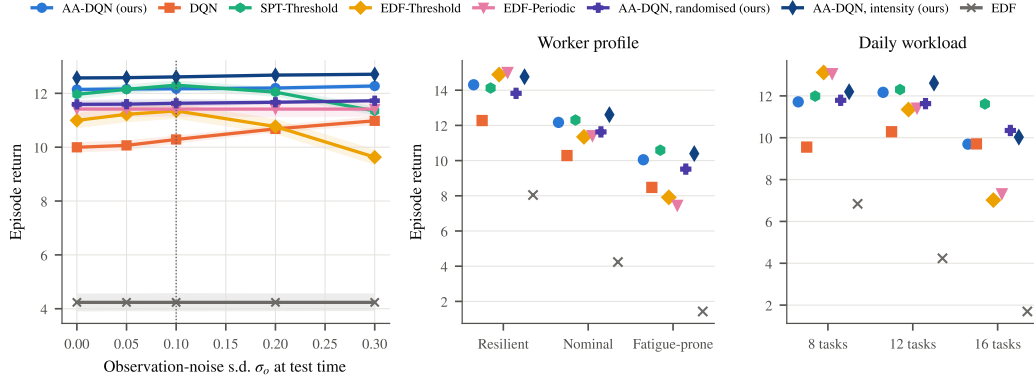


Figure 9: Zero-shot robustness. Left: return versus test-time observation noise (training/tuning at $\sigma_o = 0.1$, dotted line; shaded: 95% CIs). Centre and right: return for different worker profiles and daily workloads. Heuristics use parameters tuned in the nominal environment.

647 inference is many orders of magnitude faster than the 10-minute decision interval,
 648 so the scheduler could run on a smartphone or wearable gateway. The heuristics
 649 are cheaper still, but their grid search (up to 252 parameter combinations $\times$ 300
 650 days) must be repeated whenever the environment changes, just as the agent must
 651 be retrained.

652 7. Discussion

653 7.1. When and why does learning help?

654 The experiments support a more differentiated picture than “reinforcement
 655 learning beats heuristics”. Four observations summarise it.

656 First, *the dominant effect of cognitive dynamics is that rest becomes a first-class*
 657 *scheduling decision*. Rules that ignore capacity lose a third to two thirds of the
 658 achievable return, and every competitive policy rests in 15–20% of slots. This
 659 conclusion is robust and arguably the most important for practice: any scheduler

660 for human work should include a rest decision. Even a simple periodic or threshold
661 rule recovers most of the value.

662 Second, *well-tuned, hand-designed rules are strong competitors*, and an eval-
663 uation that omits them overstates the value of learning. Our first internal exper-
664 iments compared the agent against EDF-based rules only and suggested a clear
665 advantage. That advantage shrank once SPT-based rest rules were added (Sup-
666 plementary Section S6). The pre-specified hybrid agent ties with the best rule
667 on return. The compact intensity-only agent exceeds it by a small but consistent
668 margin (secondary analysis). A naive memoryless DQN would have led to the
669 opposite conclusion. These sensitivities echo concerns in the RL evaluation lit-
670 erature [60, 61] and in reviews of DRL scheduling [47, 49]. We recommend that
671 studies of learned schedulers include tuned composite and state-aware rules as
672 baselines.

673 Third, *what learning adds is the integration of several criteria that individ-*
674 *ual rules trade off*. SPT-based rules complete many tasks but starve demanding
675 ones, whereas EDF-based rules protect demanding tasks but complete fewer. The
676 learned policies achieve high count-based and difficulty-weighted on-time com-
677 pletion simultaneously (Fig. 5). Behaviourally, they do so by working intensively
678 while the worker is fresh, scheduling demanding tasks when attention is high,
679 filling low-attention periods with easy tasks, and concentrating rest around the
680 circadian trough (Section 6.3). Each element is intuitive, but their state-, time- and
681 deadline-dependent combination is hard to express as a rule, as the hand-crafted
682 Cog-Heuristic illustrates.

683 Fourth, *the value of learning grows with the quality of the cognitive-state*
684 *estimate*. With perfect observation, AA-DQN outperforms the best rule by more

685 than one return unit while its own fatigue level drops from 0.33 to 0.27, whereas the
686 rules gain nothing from better information. The observation history is the single
687 most important architectural component, and longer histories or recurrent encoders
688 help further. Partial observability, not the optimisation itself, is therefore the main
689 bottleneck. This points to an engineering trade-off: investments in sensing and
690 state estimation are what unlock the benefits of adaptive scheduling.

691 7.2. *Implications for engineering practice*

692 The results translate into design guidance for human-centric scheduling sys-
693 tems, such as knowledge-work assistants, workflow-management platforms and
694 supervisory controllers in human–robot cells [27, 28, 123]. (i) *Always schedule*
695 *rest*. A threshold or periodic rest rule on top of SPT-like sequencing is a strong,
696 transparent default when cognitive-state estimates are noisy or unavailable. It
697 requires only a small grid search to tune. (ii) *Use learning when state estimates*
698 *are reliable or when several criteria must be balanced*. Learned policies balance
699 throughput, protection of demanding tasks and fatigue in ways that single rules
700 cannot, and they profit from better sensors. (iii) *Make the well-being weight ex-*
701 *PLICIT*. The Pareto analysis shows that, up to a point, a higher weight on fatigue
702 improves both well-being and on-time completion. The weight should be chosen
703 deliberately with the stakeholders rather than left implicit in a heuristic’s thresh-
704 olds. (iv) *Evaluate fairly*. Common random numbers, disjoint tuning and test
705 days, tuned rule baselines and multi-seed statistics are inexpensive and change
706 conclusions.

707 7.3. *Ethical and organisational considerations*

708 A scheduler that monitors attention and fatigue and recommends when a person
709 should work on what and when to rest touches on autonomy, privacy and fairness,
710 and these issues must be designed in rather than added later [108, 112, 133]. We
711 see four requirements for responsible deployment. First, *the system should be*
712 *advisory*. Recommendations should be presented to the worker, who retains the
713 authority to accept or override them; evidence from human–robot teaming indicates
714 that the allocation of decision authority affects both efficiency and satisfaction
715 [121]. Second, *cognitive-state data are sensitive personal data*. Physiological
716 and behavioural signals should be processed with informed consent, ideally on the
717 worker’s own device, and should never be used for performance appraisal. The
718 observation model of AA-DQN needs only two scalar estimates per slot, which
719 supports such data minimisation. Third, *fairness across workers* requires that
720 policies do not systematically disadvantage people whose fatigue dynamics differ
721 from the nominal profile. Section 6.8 shows how performance changes for resilient
722 and fatigue-prone profiles, and personalisation (Section 7.5) would further reduce
723 such disparities. Fourth, *safety constraints* should bound the policy, for example by
724 guaranteeing a minimum amount of rest per period regardless of deadline pressure,
725 consistent with safe RL principles [134]. Notably, the learned policies in this
726 study already increase rest relative to deadline-driven rules. The reward weight λ
727 makes the productivity–well-being trade-off explicit and auditable (Section 6.5), so
728 stakeholders can choose it deliberately rather than leaving it implicit in a heuristic.

729 7.4. *Limitations*

730 The study has limitations that define its scope. (i) *Simulation, not human data*.
731 The cognitive model is grounded in established functional forms and its parameters

were chosen to be behaviourally plausible, but it has not been calibrated against individual time-series of attention and fatigue. The absolute numbers should therefore be interpreted comparatively. The robustness analyses mitigate, but do not remove, this concern. (ii) *Abstracted sensing*. Cognitive-state estimation is modelled as additive Gaussian noise on the true state. Real estimators can be biased, drift over the day, or fail intermittently, and they may be correlated with task type. (iii) *A single worker and a single day*. Multi-day carry-over of fatigue, sleep, and team-level interactions such as hand-offs and shared deadlines are outside the scope. (iv) *Action abstraction*. The agent chooses a difficulty class or a dispatching rule rather than an individual task. This keeps the policy compact and size-invariant but excludes policies that, for example, deliberately abandon a specific hopeless task. (v) *Algorithmic scope*. We study value-based agents with a finite history. Recurrent, model-based or policy-gradient agents [65, 127, 145] might extract more from the observation sequence. (vi) *Simulated time-on-task effects only*. Motivation, task interest, interruptions and errors, which also shape real productivity [115, 116, 146], are not modelled. (vii) *Generalisation across workloads*. Policies trained on 12 tasks per day did not transfer well to 8 or 16 tasks, and simple domain randomisation did not resolve this (Section 6.8). (viii) *Secondary analysis*. The intensity-only variant was identified as promising in a component study before its full evaluation. Although it was then evaluated with the complete ten-seed protocol on the same held-out days, its result should be confirmed on independent conditions before it is considered the preferred design.

7.5. Future work

Three extensions follow directly. First, *empirical calibration and validation*: fitting the latent dynamics to longitudinal data from vigilance probes, wearables

757 and task logs [52, 55, 104], then evaluating the scheduler in a controlled user study
758 with NASA-TLX and performance outcomes [56], ideally as a micro-randomised
759 trial [117]. Second, *personalisation*: domain randomisation over worker profiles
760 during training [147, 148], and Bayesian or meta-learning adaptation to an individ-
761 ual’s parameters, in the spirit of personalised just-in-time interventions [118, 119].
762 Third, *multi-worker and human–robot settings*: extending the formulation to teams
763 and to collaborative cells in which a robot absorbs demanding sub-tasks when a
764 co-worker’s attention is low [122–125], with constrained or multi-objective RL to
765 enforce explicit well-being guarantees [63].

766 8. Conclusion

767 This paper formulated attention-aware task scheduling as a partially observable
768 Markov decision process. In this formulation a worker’s attention, fatigue and
769 circadian state shape their capacity, rest is an explicit action, and the cognitive
770 state is observed only through noisy estimates. We proposed a memory-augmented
771 deep Q-network (AA-DQN) and evaluated it against nine rule-based schedulers,
772 including grid-tuned EDF and SPT rules with periodic or state-triggered rest,
773 using ten training seeds, 500 held-out workdays with common random numbers,
774 and multiplicity-corrected tests.

775 Rules that ignore cognitive dynamics performed poorly, and every competitive
776 policy rested in 15–20% of the day. The pre-specified hybrid AA-DQN matched
777 the return of the best tuned rule and achieved a higher on-time rate (72.8% versus
778 71.4%). A compact variant restricted to intensity actions, evaluated as a sec-
779 ondary analysis, exceeded the best rule in return (+0.31), on-time rate (74.0%) and
780 difficulty-weighted on-time rate (67.8%). The learned policies are the only ones

781 that combine high task throughput with protection of demanding tasks. They work
782 intensively while the worker is fresh, schedule hard tasks when attention is high
783 and rest around the post-lunch dip. Ablations showed that the value of learning
784 grows sharply with the reliability of cognitive-state estimates, that memory is the
785 critical architectural component, and that a moderate weight on fatigue improves
786 both well-being and productivity.

787 The study is based on a behaviourally grounded but uncalibrated simulator.
788 Its main limitation is therefore the gap to real workers, which future work will
789 close through calibration with physiological and behavioural data, personalised
790 policies, and user studies. The released simulator, agents and evaluation protocol
791 provide a reproducible basis for that work and for fair comparisons of learned and
792 hand-designed schedulers in human-centric systems.

793 **CRedit authorship contribution statement**

794 **P.N. Bhargav Teja:** Conceptualization, Methodology, Software, Formal
795 analysis, Investigation, Visualization, Writing – original draft. **Lanka Sree**
796 **Chathurya:** Methodology, Software, Validation, Data curation, Investigation,
797 Writing – original draft. **Rajay Vedaraj I.S.:** Conceptualization, Supervision,
798 Methodology, Resources, Project administration, Writing – review & editing.

799 **Declaration of competing interest**

800 The authors declare that they have no known competing financial interests or
801 personal relationships that could have appeared to influence the work reported in
802 this paper.

803 **Funding**

804 This research did not receive any specific grant from funding agencies in the
805 public, commercial, or not-for-profit sectors.

806 **Data availability**

807 The simulator, agents, baselines, experiment configurations, raw per-episode
808 results and the scripts that generate every table and figure are openly available at
809 <https://github.com/bhargavteja-9779/r1>. No human-subject data were
810 collected.

811 **Declaration of generative AI and AI-assisted technologies in the manuscript** 812 **preparation process**

813 During the preparation of this work the authors used an AI assistant (Claude,
814 Anthropic) to support code development, language editing and the organisation of
815 the manuscript. After using this tool, the authors reviewed and edited the content
816 as needed and take full responsibility for the content of the published article.

817 **Acknowledgements**

818 The authors thank Vellore Institute of Technology for providing the computa-
819 tional facilities used in this study.

820 **References**

- 821 [1] W. E. Smith, Various optimizers for single-stage production, Naval Re-
822 search Logistics Quarterly 3 (1-2) (1956) 59–66. doi:10.1002/nav.
823 3800030106.

- 824 [2] J. R. Jackson, Scheduling a production line to minimize maximum tardiness,
825 Research Report 43, Management Science Research Project, University of
826 California, Los Angeles (1955).
- 827 [3] R. Graham, E. Lawler, J. Lenstra, A. Kan, Optimization and Approximation
828 in Deterministic Sequencing and Scheduling: a Survey, Elsevier, 1979, pp.
829 287–326. doi:10.1016/s0167-5060(08)70356-x.
- 830 [4] M. L. Pinedo, Scheduling, Springer International Publishing, 2016. doi:
831 10.1007/978-3-319-26580-3.
- 832 [5] C. L. Liu, J. W. Layland, Scheduling algorithms for multiprogramming in
833 a hard-real-time environment, Journal of the ACM 20 (1) (1973) 46–61.
834 doi:10.1145/321738.321743.
- 835 [6] N. H. Mackworth, The breakdown of vigilance during prolonged visual
836 search, Quarterly Journal of Experimental Psychology 1 (1) (1948) 6–21.
837 doi:10.1080/17470214808416738.
- 838 [7] J. S. Warm, R. Parasuraman, G. Matthews, Vigilance requires hard mental
839 work and is stressful, Human Factors: The Journal of the Human Fac-
840 tors and Ergonomics Society 50 (3) (2008) 433–441. doi:10.1518/
841 001872008x312152.
- 842 [8] D. R. Thomson, D. Besner, D. Smilek, A resource-control account
843 of sustained attention: Evidence from mind-wandering and vigilance
844 paradigms, Perspectives on Psychological Science 10 (1) (2015) 82–96.
845 doi:10.1177/1745691614556681.

- 846 [9] M. Esterman, D. Rothlein, Models of sustained attention, *Current Opinion in*
847 *Psychology* 29 (2019) 174–180. doi:10.1016/j.copsyc.2019.03.005.
- 848 [10] M. A. Boksem, M. Tops, Mental fatigue: Costs and benefits, *Brain Research*
849 *Reviews* 59 (1) (2008) 125–139. doi:10.1016/j.brainresrev.2008.
850 07.001.
- 851 [11] M. M. Lorist, M. Klein, S. Nieuwenhuis, R. de Jong, G. Mulder, T. F.
852 Meijman, Mental fatigue and task control: Planning and preparation,
853 *Psychophysiology* 37 (5) (2000) 614–625. doi:10.1111/1469-8986.
854 3750614.
- 855 [12] R. Hockey, *The Psychology of Fatigue: Work, Effort and Control*, Cam-
856 bridge University Press, 2013. doi:10.1017/cbo9781139015394.
- 857 [13] P. L. Ackerman (Ed.), *Cognitive Fatigue: Multidisciplinary Perspectives on*
858 *Current Research and Future Applications*, American Psychological Asso-
859 ciation, Washington, DC, 2011. doi:10.1037/12343-000.
- 860 [14] T. Åkerstedt, S. Folkard, The three-process model of alertness and its ex-
861 tension to performance, sleep latency, and sleep length, *Chronobiology In-*
862 *ternational* 14 (2) (1997) 115–123. doi:10.3109/07420529709001149.
- 863 [15] T. H. Monk, The post-lunch dip in performance, *Clinics in Sports Medicine*
864 24 (2) (2005) e15–e23. doi:10.1016/j.csm.2004.12.002.
- 865 [16] C. Schmidt, F. Collette, C. Cajochen, P. Peigneux, A time to think: Circadian
866 rhythms in human cognition, *Cognitive Neuropsychology* 24 (7) (2007)
867 755–789. doi:10.1080/02643290701754158.

- 868 [17] K. Blatter, C. Cajochen, Circadian rhythms in cognitive performance:
869 Methodological constraints, protocols, theoretical underpinnings, Physi-
870 ology & Behavior 90 (2-3) (2007) 196–208. doi:10.1016/j.physbeh.
871 2006.09.009.
- 872 [18] R. A. HENNING, P. JACQUES, G. V. KISSEL, A. B. SULLIVAN, S. M.
873 ALTERAS-WEBB, Frequent short rest breaks from computer work: effects
874 on productivity and well-being at two field sites, Ergonomics 40 (1) (1997)
875 78–91. doi:10.1080/001401397188396.
- 876 [19] A. J. Dababneh, N. Swanson, R. L. Shell, Impact of added rest breaks on the
877 productivity and well being of workers, Ergonomics 44 (2) (2001) 164–174.
878 doi:10.1080/00140130121538.
- 879 [20] T. L. Galinsky, N. G. Swanson, S. L. Sauter, J. J. Hurrell, L. M. Schleifer,
880 A field study of supplementary rest breaks for data-entry operators, Er-
881 gonomics 43 (5) (2000) 622–638. doi:10.1080/001401300184297.
- 882 [21] P. Tucker, The impact of rest breaks upon accident risk, fatigue and
883 performance: A review, Work & Stress 17 (2) (2003) 123–137. doi:
884 10.1080/0267837031000155949.
- 885 [22] P. Albulescu, I. Macsinga, A. Rusu, C. Sulea, A. Bodnaru, B. T. Tulbure,
886 “give me a break!” a systematic review and meta-analysis on the efficacy
887 of micro-breaks for increasing well-being and performance, PLOS ONE
888 17 (8) (2022) e0272460. doi:10.1371/journal.pone.0272460.
- 889 [23] D. Dawson, K. Reid, Fatigue, alcohol and performance impairment, Nature
890 388 (6639) (1997) 235–235. doi:10.1038/40775.

- 891 [24] A. Williamson, D. A. Lombardi, S. Folkard, J. Stutts, T. K. Courtney,
892 J. L. Connor, The link between fatigue and safety, *Accident Analysis &*
893 *Prevention* 43 (2) (2011) 498–515. doi:10.1016/j.aap.2009.11.011.
- 894 [25] M. Breque, L. De Nul, A. Petridis, Industry 5.0: Towards a sustainable,
895 human-centric and resilient European industry, Tech. rep., European Com-
896 mission, Directorate-General for Research and Innovation, Luxembourg
897 (2021).
- 898 [26] X. Xu, Y. Lu, B. Vogel-Heuser, L. Wang, Industry 4.0 and industry 5.0—
899 inception, conception and perception, *Journal of Manufacturing Systems*
900 61 (2021) 530–535. doi:10.1016/j.jmsy.2021.10.006.
- 901 [27] Y. Lu, H. Zheng, S. Chand, W. Xia, Z. Liu, X. Xu, L. Wang, Z. Qin, J. Bao,
902 Outlook on human-centric manufacturing towards industry 5.0, *Journal*
903 *of Manufacturing Systems* 62 (2022) 612–627. doi:10.1016/j.jmsy.
904 2022.02.001.
- 905 [28] J. Leng, W. Sha, B. Wang, P. Zheng, C. Zhuang, Q. Liu, T. Wuest,
906 D. Mourtzis, L. Wang, Industry 5.0: Prospect and retrospect, *Journal*
907 *of Manufacturing Systems* 65 (2022) 279–295. doi:10.1016/j.jmsy.
908 2022.09.017.
- 909 [29] D. Ivanov, The industry 5.0 framework: viability-based integration of the
910 resilience, sustainability, and human-centricity perspectives, *International*
911 *Journal of Production Research* 61 (5) (2022) 1683–1695. doi:10.1080/
912 00207543.2022.2118892.

- 913 [30] M. Jaber, W. Neumann, Modelling worker fatigue and recovery in dual-
914 resource constrained systems, *Computers & Industrial Engineering* 59 (1)
915 (2010) 75–84. doi:10.1016/j.cie.2010.03.001.
- 916 [31] M. Jaber, Z. Givi, W. Neumann, Incorporating human fatigue and recovery
917 into the learning–forgetting process, *Applied Mathematical Modelling*
918 37 (12-13) (2013) 7287–7299. doi:10.1016/j.apm.2013.02.028.
- 919 [32] Z. Givi, M. Jaber, W. Neumann, Modelling worker reliability with learning
920 and fatigue, *Applied Mathematical Modelling* 39 (17) (2015) 5186–5199.
921 doi:10.1016/j.apm.2015.03.038.
- 922 [33] S. E. Bechtold, R. E. Janaro, D. W. L. Sumners, Maximization of labor
923 productivity through optimal rest-break schedules, *Management Science*
924 30 (12) (1984) 1442–1458. doi:10.1287/mnsc.30.12.1442.
- 925 [34] S. Konz, Work/rest: Part ii – the scientific basis (knowledge base) for
926 the guide, *International Journal of Industrial Ergonomics* 22 (1-2) (1998)
927 73–99. doi:10.1016/s0169-8141(97)00069-3.
- 928 [35] M. Calzavara, A. Persona, F. Sgarbossa, V. Visentin, A model for rest
929 allowance estimation to improve tasks assignment to operators, *International*
930 *Journal of Production Research* 57 (3) (2018) 948–962. doi:10.1080/
931 00207543.2018.1497816.
- 932 [36] E. J. Lodree, C. D. Geiger, X. Jiang, Taxonomy for integrating scheduling
933 theory and human factors: Review and research opportunities, *International*
934 *Journal of Industrial Ergonomics* 39 (1) (2009) 39–51. doi:10.1016/j.
935 ergon.2008.05.001.

- 936 [37] W. Patrick Neumann, J. Dul, Human factors: spanning the gap between om
937 and hrm, *International Journal of Operations & Production Management*
938 30 (9) (2010) 923–950. doi:10.1108/01443571011075056.
- 939 [38] J. Boudreau, W. Hopp, J. O. McClain, L. J. Thomas, On the interface
940 between operations and human resources management, *Manufacturing &*
941 *Service Operations Management* 5 (3) (2003) 179–202. doi:10.1287/
942 msom.5.3.179.16032.
- 943 [39] N. Katiraei, M. Calzavara, S. Finco, D. Battini, O. Battaïa, Consideration of
944 workers’ differences in production systems modelling and design: State of
945 the art and directions for future research, *International Journal of Production*
946 *Research* 59 (11) (2021) 3237–3268. doi:10.1080/00207543.2021.
947 1884766.
- 948 [40] H. Mao, M. Alizadeh, I. Menache, S. Kandula, Resource management with
949 deep reinforcement learning, in: *Proceedings of the 15th ACM Workshop*
950 *on Hot Topics in Networks, HotNets-XV*, ACM, 2016, pp. 50–56. doi:
951 10.1145/3005745.3005750.
- 952 [41] H. Mao, M. Schwarzkopf, S. B. Venkatakrisnan, Z. Meng, M. Alizadeh,
953 Learning scheduling algorithms for data processing clusters, in: *Pro-*
954 *ceedings of the ACM Special Interest Group on Data Communication,*
955 *SIGCOMM ’19*, ACM, 2019, pp. 270–288. doi:10.1145/3341302.
956 3342080.
- 957 [42] B. Waschneck, A. Reichstaller, L. Belzner, T. Altenmüller, T. Bauernhansl,
958 A. Knapp, A. Kyek, Optimization of global production scheduling with

- 959 deep reinforcement learning, *Procedia CIRP* 72 (2018) 1264–1269. doi:
960 10.1016/j.procir.2018.03.212.
- 961 [43] S. Luo, Dynamic scheduling for flexible job shop with new job insertions by
962 deep reinforcement learning, *Applied Soft Computing* 91 (2020) 106208.
963 doi:10.1016/j.asoc.2020.106208.
- 964 [44] C. Zhang, W. Song, Z. Cao, J. Zhang, P. S. Tan, C. Xu, Learning to dispatch
965 for job shop scheduling via deep reinforcement learning, in: *Advances in*
966 *Neural Information Processing Systems*, Vol. 33, 2020, pp. 1621–1632.
- 967 [45] J. Park, J. Chun, S. H. Kim, Y. Kim, J. Park, Learning to schedule job-shop
968 problems: representation and policy learning using graph neural network
969 and reinforcement learning, *International Journal of Production Research*
970 59 (11) (2021) 3360–3377. doi:10.1080/00207543.2020.1870013.
- 971 [46] W. Song, X. Chen, Q. Li, Z. Cao, Flexible job-shop scheduling via graph
972 neural network and deep reinforcement learning, *IEEE Transactions on In-*
973 *dustrial Informatics* 19 (2) (2023) 1600–1610. doi:10.1109/tii.2022.
974 3189725.
- 975 [47] B. M. Kayhan, G. Yildiz, Reinforcement learning applications to ma-
976 chine scheduling problems: a comprehensive literature review, *Journal*
977 *of Intelligent Manufacturing* 34 (3) (2021) 905–929. doi:10.1007/
978 s10845-021-01847-3.
- 979 [48] M. Panzer, B. Bender, Deep reinforcement learning in production systems:
980 a systematic literature review, *International Journal of Production Research*
981 60 (13) (2021) 4316–4341. doi:10.1080/00207543.2021.1973138.

- 982 [49] M. Khadivi, T. Charter, M. Yaghoubi, M. Jalayer, M. Ahang, A. Sho-
 983 jaeinasab, H. Najjaran, Deep reinforcement learning for machine schedul-
 984 ing: Methodology, the state-of-the-art, and future directions, *Computers &*
 985 *Industrial Engineering* 200 (2025) 110856. doi:10.1016/j.cie.2025.
 986 110856.
- 987 [50] W. Tan, X. Yuan, J. Wang, X. Zhang, A fatigue-conscious dual resource
 988 constrained flexible job shop scheduling problem by enhanced nsga-ii: An
 989 application from casting workshop, *Computers & Industrial Engineering*
 990 160 (2021) 107557. doi:10.1016/j.cie.2021.107557.
- 991 [51] Y. Hu, Z. Zhang, J. Chen, D. Tang, Q. Cai, Graph-based deep reinforcement
 992 learning for dynamic scheduling of flexible job-shop considering worker
 993 fatigue and multi-skill factors, *Applied Soft Computing* 184 (2025) 113712.
 994 doi:10.1016/j.asoc.2025.113712.
- 995 [52] R. L. Charles, J. Nixon, Measuring mental workload using physiological
 996 measures: A systematic review, *Applied Ergonomics* 74 (2019) 221–232.
 997 doi:10.1016/j.apergo.2018.08.028.
- 998 [53] G. Borghini, L. Astolfi, G. Vecchiato, D. Mattia, F. Babiloni, Measuring
 999 neurophysiological signals in aircraft pilots and car drivers for the assess-
 1000 ment of mental workload, fatigue and drowsiness, *Neuroscience & Biobe-*
 1001 *havioral Reviews* 44 (2014) 58–75. doi:10.1016/j.neubiorev.2012.
 1002 10.003.
- 1003 [54] M. A. Hogervorst, A.-M. Brouwer, J. B. F. van Erp, Combining and com-
 1004 paring eeg, peripheral physiology and eye-related measures for the as-

- 1005 sessment of mental workload, *Frontiers in Neuroscience* 8 (Oct. 2014).
1006 doi:10.3389/fnins.2014.00322.
- 1007 [55] M. Basner, D. F. Dinges, Maximizing sensitivity of the psychomotor
1008 vigilance test (pvt) to sleep loss, *Sleep* 34 (5) (2011) 581–591. doi:
1009 10.1093/sleep/34.5.581.
- 1010 [56] S. G. Hart, L. E. Staveland, Development of NASA-TLX (task load in-
1011 dex): Results of empirical and theoretical research, in: P. A. Han-
1012 cock, N. Meshkati (Eds.), *Human Mental Workload*, Vol. 52 of *Ad-
1013 vances in Psychology*, North-Holland, Amsterdam, 1988, pp. 139–183.
1014 doi:10.1016/s0166-4115(08)62386-9.
- 1015 [57] K. Åström, Optimal control of markov processes with incomplete state
1016 information, *Journal of Mathematical Analysis and Applications* 10 (1)
1017 (1965) 174–205. doi:10.1016/0022-247x(65)90154-x.
- 1018 [58] R. D. Smallwood, E. J. Sondik, The optimal control of partially observable
1019 markov processes over a finite horizon, *Operations Research* 21 (5) (1973)
1020 1071–1088. doi:10.1287/opre.21.5.1071.
- 1021 [59] L. P. Kaelbling, M. L. Littman, A. R. Cassandra, Planning and acting in
1022 partially observable stochastic domains, *Artificial Intelligence* 101 (1-2)
1023 (1998) 99–134. doi:10.1016/s0004-3702(98)00023-x.
- 1024 [60] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, D. Meger, Deep
1025 reinforcement learning that matters, *Proceedings of the AAAI Conference
1026 on Artificial Intelligence* 32 (1) (Apr. 2018). doi:10.1609/aaai.v32i1.
1027 11694.

- 1028 [61] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare,
1029 Deep reinforcement learning at the edge of the statistical precipice, in:
1030 Advances in Neural Information Processing Systems, Vol. 34, 2021, pp.
1031 29304–29320.
- 1032 [62] D. M. Roijers, P. Vamplew, S. Whiteson, R. Dazeley, A survey of multi-
1033 objective sequential decision-making, Journal of Artificial Intelligence Re-
1034 search 48 (2013) 67–113. doi:10.1613/jair.3987.
- 1035 [63] C. F. Hayes, R. Rădulescu, E. Bargiacchi, J. Källström, M. Macfarlane,
1036 M. Reymond, T. Verstraeten, L. M. Zintgraf, R. Dazeley, F. Heintz, E. How-
1037 ley, A. A. Irissappane, P. Mannion, A. Nowé, G. Ramos, M. Restelli,
1038 P. Vamplew, D. M. Roijers, A practical guide to multi-objective reinforce-
1039 ment learning and planning, Autonomous Agents and Multi-Agent Systems
1040 36 (1) (Apr. 2022). doi:10.1007/s10458-022-09552-y.
- 1041 [64] S. P. Singh, T. Jaakkola, M. I. Jordan, Learning Without State-Estimation
1042 in Partially Observable Markovian Decision Processes, Elsevier, 1994, pp.
1043 284–292. doi:10.1016/b978-1-55860-335-6.50042-8.
- 1044 [65] M. Hausknecht, P. Stone, Deep recurrent Q-learning for partially observable
1045 MDPs, in: AAAI Fall Symposium on Sequential Decision Making for
1046 Intelligent Agents, 2015, pp. 29–37.
- 1047 [66] T. Ni, B. Eysenbach, R. Salakhutdinov, Recurrent model-free RL can be a
1048 strong baseline for many POMDPs, in: Proceedings of the 39th International
1049 Conference on Machine Learning, Vol. 162 of PMLR, 2022, pp. 16691–
1050 16723.

- 1051 [67] H. Van Hasselt, A. Guez, D. Silver, Deep reinforcement learning with double
1052 q-learning, Proceedings of the AAAI Conference on Artificial Intelligence
1053 30 (1) (Mar. 2016). doi:10.1609/aaai.v30i1.10295.
- 1054 [68] Z. Wang, T. Schaul, M. Hessel, H. van Hasselt, M. Lanctot, N. de Freitas,
1055 Dueling network architectures for deep reinforcement learning, in: Proceed-
1056 ings of the 33rd International Conference on Machine Learning, Vol. 48 of
1057 PMLR, 2016, pp. 1995–2003.
- 1058 [69] J. M. Moore, An n job, one machine sequencing algorithm for minimizing
1059 the number of late jobs, Management Science 15 (1) (1968) 102–109.
1060 doi:10.1287/mnsc.15.1.102.
- 1061 [70] G. C. Buttazzo, Hard Real-Time Computing Systems: Predictable Schedul-
1062 ing Algorithms and Applications, Springer US, 2011. doi:10.1007/
1063 978-1-4614-0676-1.
- 1064 [71] S. S. Panwalkar, W. Iskander, A survey of scheduling rules, Operations
1065 Research 25 (1) (1977) 45–61. doi:10.1287/opre.25.1.45.
- 1066 [72] R. Haupt, A survey of priority rule-based scheduling, OR Spektrum 11 (1)
1067 (1989) 3–16. doi:10.1007/bf01721162.
- 1068 [73] G. E. Vieira, J. W. Herrmann, E. Lin, Rescheduling manufacturing systems:
1069 A framework of strategies, policies, and methods, Journal of Scheduling
1070 6 (1) (2003) 39–62. doi:10.1023/a:1022235519958.
- 1071 [74] D. Ouelhadj, S. Petrovic, A survey of dynamic scheduling in manufacturing
1072 systems, Journal of Scheduling 12 (4) (2008) 417–431. doi:10.1007/
1073 s10951-008-0090-8.

- 1074 [75] A. Ernst, H. Jiang, M. Krishnamoorthy, D. Sier, Staff scheduling and
1075 rostering: A review of applications, methods and models, *European*
1076 *Journal of Operational Research* 153 (1) (2004) 3–27. doi:10.1016/
1077 s0377-2217(03)00095-x.
- 1078 [76] E. K. Burke, P. De Causmaecker, G. V. Berghe, H. Van Landeghem, The
1079 state of the art of nurse rostering, *Journal of Scheduling* 7 (6) (2004) 441–
1080 499. doi:10.1023/b:josh.00000046076.75950.0b.
- 1081 [77] J. Van den Bergh, J. Beliën, P. De Bruecker, E. Demeulemeester,
1082 L. De Boeck, Personnel scheduling: A literature review, *European Jour-*
1083 *nal of Operational Research* 226 (3) (2013) 367–385. doi:10.1016/j.
1084 ejor.2012.11.029.
- 1085 [78] T. P. WRIGHT, Factors affecting the cost of airplanes, *Journal of the Aero-*
1086 *nautical Sciences* 3 (4) (1936) 122–128. doi:10.2514/8.155.
- 1087 [79] E. H. Grosse, C. H. Glock, W. P. Neumann, Human factors in order pick-
1088 ing: a content analysis of the literature, *International Journal of Produc-*
1089 *tion Research* 55 (5) (2016) 1260–1276. doi:10.1080/00207543.2016.
1090 1186296.
- 1091 [80] B. A. Kadir, O. Broberg, C. S. d. Conceição, Current research and future
1092 perspectives on human factors and ergonomics in industry 4.0, *Computers &*
1093 *Industrial Engineering* 137 (2019) 106004. doi:10.1016/j.cie.2019.
1094 106004.
- 1095 [81] W. P. Neumann, S. Winkelhaus, E. H. Grosse, C. H. Glock, Industry 4.0
1096 and the human factor – a systems framework and analysis methodology

- 1097 for successful development, *International Journal of Production Economics*
1098 233 (2021) 107992. doi:10.1016/j.ijpe.2020.107992.
- 1099 [82] W. Zhang, T. G. Dietterich, A reinforcement learning approach to job-shop
1100 scheduling, in: *Proceedings of the 14th International Joint Conference on*
1101 *Artificial Intelligence*, 1995, pp. 1114–1120.
- 1102 [83] M. Aydin, E. Öztemel, Dynamic job-shop scheduling using reinforcement
1103 learning agents, *Robotics and Autonomous Systems* 33 (2-3) (2000) 169–
1104 178. doi:10.1016/s0921-8890(00)00087-7.
- 1105 [84] C. J. C. H. Watkins, P. Dayan, Q-learning, *Machine Learning* 8 (3-4) (1992)
1106 279–292. doi:10.1007/bf00992698.
- 1107 [85] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G.
1108 Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski,
1109 S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Ku-
1110 maran, D. Wierstra, S. Legg, D. Hassabis, Human-level control through
1111 deep reinforcement learning, *Nature* 518 (7540) (2015) 529–533. doi:
1112 10.1038/nature14236.
- 1113 [86] C.-L. Liu, C.-C. Chang, C.-J. Tseng, Actor-critic deep reinforcement learn-
1114 ing for solving job shop scheduling problems, *IEEE Access* 8 (2020) 71752–
1115 71762. doi:10.1109/access.2020.2987820.
- 1116 [87] J. Shahrabi, M. A. Adibi, M. Mahootchi, A reinforcement learning approach
1117 to parameter estimation in dynamic job shop scheduling, *Computers &*
1118 *Industrial Engineering* 110 (2017) 75–82. doi:10.1016/j.cie.2017.
1119 05.026.

- 1120 [88] Y. Bengio, A. Lodi, A. Prouvost, Machine learning for combinatorial op-
1121 timization: A methodological tour d’horizon, *European Journal of Oper-*
1122 *ational Research* 290 (2) (2021) 405–421. doi:10.1016/j.ejor.2020.
1123 07.063.
- 1124 [89] N. Mazyavkina, S. Sviridov, S. Ivanov, E. Burnaev, Reinforcement learn-
1125 ing for combinatorial optimization: A survey, *Computers & Operations*
1126 *Research* 134 (2021) 105400. doi:10.1016/j.cor.2021.105400.
- 1127 [90] D. Kahneman, *Attention and Effort*, Prentice-Hall, Englewood Cliffs, NJ,
1128 1973.
- 1129 [91] C. D. Wickens, Multiple resources and mental workload, *Human Factors:*
1130 *The Journal of the Human Factors and Ergonomics Society* 50 (3) (2008)
1131 449–455. doi:10.1518/001872008x288394.
- 1132 [92] J. Sweller, Cognitive load during problem solving: Effects on learn-
1133 ing, *Cognitive Science* 12 (2) (1988) 257–285. doi:10.1207/
1134 s15516709cog1202_4.
- 1135 [93] A. Ariga, A. Lleras, Brief and rare mental “breaks” keep you focused:
1136 Deactivation and reactivation of task goals preempt vigilance decrements,
1137 *Cognition* 118 (3) (2011) 439–443. doi:10.1016/j.cognition.2010.
1138 12.007.
- 1139 [94] R. Kurzban, A. Duckworth, J. W. Kable, J. Myers, An opportunity cost
1140 model of subjective effort and task performance, *Behavioral and Brain*
1141 *Sciences* 36 (6) (2013) 661–679. doi:10.1017/s0140525x12003196.

- 1142 [95] J. F. Hopstaken, D. van der Linden, A. B. Bakker, M. A. J. Kompier,
1143 A multifaceted investigation of the link between mental fatigue and task
1144 disengagement, *Psychophysiology* 52 (3) (2014) 305–315. doi:10.1111/
1145 psyp.12339.
- 1146 [96] A. A. Borbély, A two process model of sleep regulation, *Human Neurobi-*
1147 *ology* 1 (3) (1982) 195–204.
- 1148 [97] S. Daan, D. G. Beersma, A. A. Borbely, Timing of human sleep: recovery
1149 process gated by a circadian pacemaker, *American Journal of Physiology-*
1150 *Regulatory, Integrative and Comparative Physiology* 246 (2) (1984) R161–
1151 R183. doi:10.1152/ajpregu.1984.246.2.r161.
- 1152 [98] A. A. Borbély, S. Daan, A. Wirz-Justice, T. Deboer, The two-process model
1153 of sleep regulation: a reappraisal, *Journal of Sleep Research* 25 (2) (2016)
1154 131–143. doi:10.1111/jsr.12371.
- 1155 [99] S. R. Hursh, D. P. Redmond, M. L. Johnson, D. R. Thorne, G. Belenky, T. J.
1156 Balkin, W. F. Storm, J. C. Miller, D. R. Eddy, Fatigue models for applied re-
1157 search in warfighting, *Aviation, Space, and Environmental Medicine* 75 (3,
1158 Suppl.) (2004) A44–A53.
- 1159 [100] S. Sonnentag, C. Fritz, The recovery experience questionnaire: Develop-
1160 ment and validation of a measure for assessing recuperation and unwinding
1161 from work., *Journal of Occupational Health Psychology* 12 (3) (2007) 204–
1162 221. doi:10.1037/1076-8998.12.3.204.
- 1163 [101] C. Fritz, A. M. Ellis, C. A. Demsky, B. C. Lin, F. Guros, Embracing work

- breaks, *Organizational Dynamics* 42 (4) (2013) 274–280. doi:10.1016/j.orgdyn.2013.07.005.
- [102] S. Kim, Y. Park, Q. Niu, Micro-break activities at work to recover from daily work demands, *Journal of Organizational Behavior* 38 (1) (2016) 28–44. doi:10.1002/job.2109.
- [103] A. Gevins, M. E. Smith, Neurophysiological measures of cognitive workload during human-computer interaction, *Theoretical Issues in Ergonomics Science* 4 (1-2) (2003) 113–131. doi:10.1080/14639220210159717.
- [104] F. Shaffer, J. P. Ginsberg, An overview of heart rate variability metrics and norms, *Frontiers in Public Health* 5 (2017). doi:10.3389/fpubh.2017.00258.
- [105] D. F. Dinges, J. W. Powell, Microcomputer analyses of performance on a portable, simple visual rt task during sustained operations, *Behavior Research Methods, Instruments, & Computers* 17 (6) (1985) 652–655. doi:10.3758/bf03200977.
- [106] M. S. Young, K. A. Brookhuis, C. D. Wickens, P. A. Hancock, State of science: mental workload in ergonomics, *Ergonomics* 58 (1) (2014) 1–17. doi:10.1080/00140139.2014.956151.
- [107] E. A. Byrne, R. Parasuraman, Psychophysiology and adaptive automation, *Biological Psychology* 42 (3) (1996) 249–268. doi:10.1016/0301-0511(95)05161-9.
- [108] R. Parasuraman, T. Sheridan, C. Wickens, A model for types and levels of human interaction with automation, *IEEE Transactions on Systems, Man,*

- 1187 and Cybernetics - Part A: Systems and Humans 30 (3) (2000) 286–297.
1188 doi:10.1109/3468.844354.
- 1189 [109] D. B. Kaber, M. R. Endsley, The effects of level of automation and adaptive
1190 automation on human performance, situation awareness and workload in
1191 a dynamic control task, Theoretical Issues in Ergonomics Science 5 (2)
1192 (2004) 113–153. doi:10.1080/1463922021000054335.
- 1193 [110] S. H. Fairclough, Fundamentals of physiological computing, Interacting
1194 with Computers 21 (1-2) (2009) 133–145. doi:10.1016/j.intcom.
1195 2008.10.011.
- 1196 [111] T. O. Zander, C. Kothe, Towards passive brain–computer interfaces: ap-
1197 plying brain–computer interface technology to human–machine systems
1198 in general, Journal of Neural Engineering 8 (2) (2011) 025005. doi:
1199 10.1088/1741-2560/8/2/025005.
- 1200 [112] K. M. Feigh, M. C. Dorneich, C. C. Hayes, Toward a characterization
1201 of adaptive systems: A framework for researchers and system designers,
1202 Human Factors: The Journal of the Human Factors and Ergonomics Society
1203 54 (6) (2012) 1008–1024. doi:10.1177/0018720812443983.
- 1204 [113] E. Horvitz, J. Apacible, Learning and reasoning about interruption, in:
1205 Proceedings of the 5th international conference on Multimodal interfaces,
1206 ICMI-PUI03, ACM, 2003, pp. 20–27. doi:10.1145/958432.958440.
- 1207 [114] S. T. Iqbal, B. P. Bailey, Effects of intelligent notification management
1208 on users and their tasks, in: Proceedings of the SIGCHI Conference on

- 1209 Human Factors in Computing Systems, CHI '08, ACM, 2008, pp. 93–102.
1210 doi:10.1145/1357054.1357070.
- 1211 [115] M. Czerwinski, E. Horvitz, S. Willhite, A diary study of task switching
1212 and interruptions, in: Proceedings of the SIGCHI Conference on Human
1213 Factors in Computing Systems, CHI04, ACM, 2004, pp. 175–182. doi:
1214 10.1145/985692.985715.
- 1215 [116] G. Mark, D. Gudith, U. Klocke, The cost of interrupted work: more speed
1216 and stress, in: Proceedings of the SIGCHI Conference on Human Factors in
1217 Computing Systems, CHI '08, ACM, 2008, pp. 107–110. doi:10.1145/
1218 1357054.1357072.
- 1219 [117] P. Klasnja, E. B. Hekler, S. Shiffman, A. Boruvka, D. Almirall, A. Tewari,
1220 S. A. Murphy, Microrandomized trials: An experimental design for devel-
1221 oping just-in-time adaptive interventions., Health Psychology 34 (Suppl)
1222 (2015) 1220–1228. doi:10.1037/hea0000305.
- 1223 [118] I. Nahum-Shani, S. N. Smith, B. J. Spring, L. M. Collins, K. Witkiewitz,
1224 A. Tewari, S. A. Murphy, Just-in-time adaptive interventions (jitais) in
1225 mobile health: Key components and design principles for ongoing health
1226 behavior support, Annals of Behavioral Medicine 52 (6) (2017) 446–462.
1227 doi:10.1007/s12160-016-9830-8.
- 1228 [119] P. Liao, K. Greenewald, P. Klasnja, S. Murphy, Personalized heartsteps:
1229 A reinforcement learning algorithm for optimizing physical activity, Pro-
1230 ceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous
1231 Technologies 4 (1) (2020) 1–22. doi:10.1145/3381007.

- 1232 [120] C. Yu, J. Liu, S. Nemati, G. Yin, Reinforcement learning in healthcare:
1233 A survey, *ACM Computing Surveys* 55 (1) (2021) 1–36. doi:10.1145/
1234 3477600.
- 1235 [121] M. C. Gombolay, R. A. Gutierrez, S. G. Clarke, G. F. Sturla, J. A. Shah,
1236 Decision-making authority, team efficiency and human worker satisfaction
1237 in mixed human–robot teams, *Autonomous Robots* 39 (3) (2015) 293–312.
1238 doi:10.1007/s10514-015-9457-9.
- 1239 [122] L. Peternel, N. Tsagarakis, D. Caldwell, A. Ajoudani, Robot adaptation
1240 to human physical fatigue in human–robot co-manipulation, *Autonomous*
1241 *Robots* 42 (5) (2017) 1011–1021. doi:10.1007/s10514-017-9678-1.
- 1242 [123] A. Ajoudani, A. M. Zanchettin, S. Ivaldi, A. Albu-Schäffer, K. Ko-
1243 suge, O. Khatib, Progress and prospects of the human–robot collab-
1244 oration, *Autonomous Robots* 42 (5) (2017) 957–975. doi:10.1007/
1245 s10514-017-9677-2.
- 1246 [124] E. Lamon, A. De Franco, L. Peternel, A. Ajoudani, A capability-aware
1247 role allocation approach to industrial assembly tasks, *IEEE Robotics and*
1248 *Automation Letters* 4 (4) (2019) 3378–3385. doi:10.1109/lra.2019.
1249 2926963.
- 1250 [125] R. Zhang, Q. Lv, J. Li, J. Bao, T. Liu, S. Liu, A reinforcement learning
1251 method for human-robot collaboration in assembly tasks, *Robotics and*
1252 *Computer-Integrated Manufacturing* 73 (2022) 102227. doi:10.1016/j.
1253 rcim.2021.102227.

- 1254 [126] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Computa-*
1255 *tion* 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.
- 1256 [127] M. Igl, L. Zintgraf, T. A. Le, F. Wood, S. Whiteson, Deep variational rein-
1257 *forcement learning for POMDPs*, in: *Proceedings of the 35th International*
1258 *Conference on Machine Learning*, Vol. 80 of PMLR, 2018, pp. 2117–2126.
- 1259 [128] L.-J. Lin, Self-improving reactive agents based on reinforcement learning,
1260 *planning and teaching*, *Machine Learning* 8 (3-4) (1992) 293–321. doi:
1261 10.1007/bf00992699.
- 1262 [129] M. Hessel, J. Modayil, H. Van Hasselt, T. Schaul, G. Ostrovski, W. Dabney,
1263 D. Horgan, B. Piot, M. Azar, D. Silver, Rainbow: Combining improvements
1264 *in deep reinforcement learning*, *Proceedings of the AAAI Conference on*
1265 *Artificial Intelligence* 32 (1) (Apr. 2018). doi:10.1609/aaai.v32i1.
1266 11796.
- 1267 [130] J. Tsitsiklis, B. Van Roy, An analysis of temporal-difference learning with
1268 *function approximation*, *IEEE Transactions on Automatic Control* 42 (5)
1269 (1997) 674–690. doi:10.1109/9.580874.
- 1270 [131] P. Vamplew, R. Dazeley, A. Berry, R. Issabekov, E. Dekker, Empir-
1271 *ical evaluation methods for multiobjective reinforcement learning al-*
1272 *gorithms*, *Machine Learning* 84 (1-2) (2010) 51–80. doi:10.1007/
1273 s10994-010-5232-5.
- 1274 [132] A. Y. Ng, D. Harada, S. Russell, Policy invariance under reward transfor-
1275 *mations: Theory and application to reward shaping*, in: *Proceedings of the*
1276 *16th International Conference on Machine Learning*, 1999, pp. 278–287.

- 1277 [133] D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, D. Mané,
1278 Concrete problems in AI safety, arXiv preprint arXiv:1606.06565 (2016).
- 1279 [134] J. García, F. Fernández, A comprehensive survey on safe reinforcement
1280 learning, *Journal of Machine Learning Research* 16 (42) (2015) 1437–1480.
- 1281 [135] G. Dulac-Arnold, N. Levine, D. J. Mankowitz, J. Li, C. Paduraru, S. Goyal,
1282 T. Hester, Challenges of real-world reinforcement learning: definitions,
1283 benchmarks and analysis, *Machine Learning* 110 (9) (2021) 2419–2468.
1284 doi:10.1007/s10994-021-05961-4.
- 1285 [136] H. P. Van Dongen, G. Maislin, J. M. Mullington, D. F. Dinges, The cu-
1286 mulative cost of additional wakefulness: Dose-response effects on neu-
1287 robehavioral functions and sleep physiology from chronic sleep restric-
1288 tion and total sleep deprivation, *Sleep* 26 (2) (2003) 117–126. doi:
1289 10.1093/sleep/26.2.117.
- 1290 [137] M. L. Puterman, *Markov Decision Processes: Discrete Stochastic Dynamic*
1291 *Programming*, Wiley, 1994. doi:10.1002/9780470316887.
- 1292 [138] R. S. Sutton, A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd
1293 Edition, MIT Press, Cambridge, MA, 2018.
- 1294 [139] P. J. Huber, Robust estimation of a location parameter, *The Annals of*
1295 *Mathematical Statistics* 35 (1) (1964) 73–101. doi:10.1214/aoms/
1296 1177703732.
- 1297 [140] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in:
1298 *International Conference on Learning Representations*, 2015.

- 1299 [141] B. Efron, Bootstrap methods: Another look at the jackknife, *The Annals of*
1300 *Statistics* 7 (1) (Jan. 1979). doi:10.1214/aos/1176344552.
- 1301 [142] F. Wilcoxon, Individual comparisons by ranking methods, *Biometrics Bul-*
1302 *letin* 1 (6) (1945) 80. doi:10.2307/3001968.
- 1303 [143] S. Holm, A simple sequentially rejective multiple test procedure, *Scandina-*
1304 *vian Journal of Statistics* 6 (2) (1979) 65–70.
- 1305 [144] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, et al.,
1306 *PyTorch: An imperative style, high-performance deep learning library*, in:
1307 *Advances in Neural Information Processing Systems*, Vol. 32, 2019, pp.
1308 8024–8035.
- 1309 [145] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal
1310 policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017).
- 1311 [146] V. M. González, G. Mark, “constant, constant, multi-tasking craziness”:
1312 managing multiple working spheres, in: *Proceedings of the SIGCHI Con-*
1313 *ference on Human Factors in Computing Systems*, CHI04, ACM, 2004, pp.
1314 113–120. doi:10.1145/985692.985707.
- 1315 [147] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, P. Abbeel, Domain
1316 randomization for transferring deep neural networks from simulation to
1317 the real world, in: *2017 IEEE/RSJ International Conference on Intelligent*
1318 *Robots and Systems (IROS)*, IEEE, 2017, pp. 23–30. doi:10.1109/iros.
1319 2017.8202133.
- 1320 [148] W. Zhao, J. P. Queralta, T. Westerlund, Sim-to-real transfer in deep re-
1321 inforcement learning for robotics: a survey, in: *2020 IEEE Symposium*

1322 Series on Computational Intelligence (SSCI), IEEE, 2020, pp. 737–744.
1323 doi:10.1109/ssci47803.2020.9308468.